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DEPARTMENT OF PHYSICS

FSK 116 / FSK 176

Laboratory Manual

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Guidelines for writing reports

A report has to be written on every experiment in this laboratory book.

Pay attention to grammatical correctness and do not mix languages. Marks will be awarded for neatness and logical presentation of your data.

i. Heading

Use the three-letter code and description of the specific experiment as the heading.

ii. Objective/Experimental Aim

Briefly state the objectives of the experiment.

iii. Results

It is not necessary to describe the theory, experimental set up and method! Only your observations, results and conclusions should be reported in a systematic and organized way.

Give all your results and wherever possible, make use of tables and graphs, **even when not specifically requested to do so**. All graphs must be drawn according to the hints in your practical manual - see experiment 2.

Always take the **accuracy** with which your **measurements** were taken into account and round your answers off to the **correct number of significant figures!**

iv. Discussion and Conclusions

A brief discussion of your observations and results, even when not requested, is required. A few points to consider are:

- Are your results in agreement with the theory?
 - What are the reasons for any observed deviation from the ideal result or behavior?
 - Can you think of experimental errors which could have influenced your results?
 - Is it possible to estimate the effect of experimental errors on your observations and results?
 - Do you have suggestions on how the results could have been improved? Do you have any suggestions or own theories?
-

Units

Every measurement of a physical quantity must be expressed in the applicable SI unit. The SI system is based on the units given in the table below. It then follows that although measurements in an experiment should be given in the units of the instrument, it is necessary to convert these units to SI units before they are used in calculations.

S.I Units

The international system of units (Système International d'Unités) is based on the mksa (meter- kilogram-second-ampere) system. The fundamental quantities are:

Quantity	Variable	Unit symbol	Unit
<i>length</i>	<i>l</i>	m	metre
<i>mass</i>	<i>m</i>	kg	kilogram
<i>time</i>	<i>t</i>	s	second
<i>electric current</i>	<i>I</i>	A	ampere
<i>temperature</i>	<i>T</i>	K	kelvin
<i>luminous intensity</i>	<i>L</i>	cd	candela
<i>amount of substance</i>	<i>n</i>	mol	mole

All other quantities can be defined in terms of the fundamental quantities.

Quantity	Variable	Unit Symbol	Unit
<i>area</i>	<i>A</i>	m ²	square metre
<i>volume</i>	<i>V</i>	m ³	cubic metre
<i>frequency</i>	<i>F</i>	Hz	hertz
<i>density</i>	ρ	kg/m ³	kilogram per cubic metre
<i>speed, velocity</i>	<i>v</i>	m/s	metre per second
<i>angular velocity</i>	ξ	rad/s	radian per second
<i>acceleration</i>	<i>a</i>	m/s ²	metre per second squared
<i>angular acceleration</i>	α	rad/s ²	radian per second squared
<i>force</i>	<i>F</i>	N	newton
<i>pressure</i>	<i>p</i>	Pa	pascal
<i>work, energy, quantity of heat</i>	<i>W</i>	J	joule
<i>power</i>	<i>P</i>	W	watt
<i>electric charge</i>	<i>Q</i>	C	coulomb
<i>potential difference, electromotive force</i>	<i>V</i>	V	volt
<i>voltage</i>	<i>V</i>	V	volt
<i>electrical resistance</i>	<i>R</i>	ϕ	ohm
<i>capacitance</i>	<i>C</i>	F	farad
<i>inductance</i>	<i>L</i>	H	henry

Multiple and sub-units

<i>Factor</i>	10^9	10^6	10^3	10^{-3}	10^{-6}	10^{-9}
<i>Prefix</i>	giga	mega	kilo	milli	micro	nano
<i>Symbol</i>	G	M	k	m	μ	n

Errors

Every measurement contains an amount of uncertainty in it, that is, no measurement is 100% accurate. This is referred to as the error in a measurement. In the first place, the measuring instrument itself sets a limit on the how accurate a reading can be. A ruler for example, that is subdivided into centimeters, will measure the length of an object correctly only to the nearest centimeter. Its error will therefore be 0.5 cm (otherwise the measurement would have been estimated to the other centimeter mark). This type of error is called the **instrumental error**. Apart from instrumental errors we also have observational and personal errors that come into play - especially the errors of parallax and interpolation estimations. The determination of the error of a measurement is so important that one should supply both the measurement and an estimation of the error of that measurement, for example the diameter of a cylinder was measured as

$$d = 4.40 \pm 0.05 \text{ cm}$$

The 0.05 cm is called the **absolute error**. We could also write

$$d = 4.40 (1 \pm 0.05/4.40) \text{ cm}$$

where 0.05/4.40 is called the **relative error** of the measurement.

The **percentage error** is expressed as a percentage of the actual value:

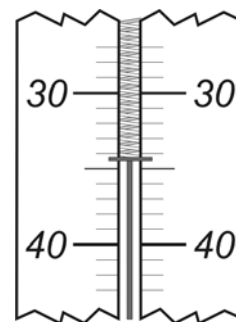
$$\% \text{ error} = \frac{\text{measured value} - \text{actual value}}{\text{actual value}} \times 100\%$$

There are methods to determine the error of a result. It is however, a difficult subject and for our purposes it is adequate to round the result off to the appropriate amount of **significant figures**.

Significant figures

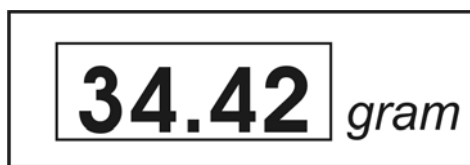
Reading an analogue scale

When taking a reading from an analogue scale you should estimate the least significant digit of the measurement by mentally subdividing the smallest division on the scale into more intervals. The diagram depicts part of a spring scale from which an object is suspended. The scale indicates the mass of the object in grams. One would report the measurement as 34.4 gram in this case. Note that the final digit of the measurement (4/10 of a gram) was estimated and has an (not too unrealistic) amount of uncertainty. The suspended object does have a definite mass which we refer to as the true mass of the object. The measuring device attempts to determine this value. The accuracy of such a measurement is the amount by which the measurement differs from the true value. If the pointer of the spring scale came to rest exactly on the 34 gram mark on the scale one should report the mass as 34.0 gram. In such an instance the final digit would still have a certain measure of uncertainty. This reflects the situation that a measurement is only an approximation of the true value being measured. For analogue scales the measurement therefore should have one more digit than that given by the smallest division on the scale. Note that readings from a graph also entail reading an analogue scale.

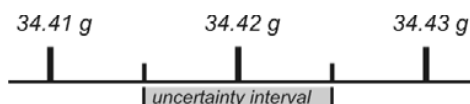


Reading a digital scale

Suppose that we weigh the same object on an electronic scale with a resolution of 0.01 gram and the scale indicates its mass as in the diagram. Even in this case there is still a measure of uncertainty in the measurement since the **true mass, m** , of the object can be anywhere within the range



$$34.415 < m < 34.425$$



and the scale would still report the mass as 34.42 grams. The true mass of the object can be anywhere in the indicated interval according to the measurement. That is the mass can be reported as 34.420 ± 0.005 grams. In short it should be reported as 34.420 gram. So, in this case we also add an additional significant digit to the measurement albeit a zero in all cases for a

digital scale. The uncertainty will again be in the least significant digit of the measurement. This final zero is therefore one digit beyond the resolution of the instrument.

When are digits significant?

- All non-zero digits are significant, for example 27 N has two significant figures and 3.46 N has three significant figures.

When zeros appear in measurements one needs to consider their position in the value in order to decide whether they are significant.

- Leading zeros are not significant, they just serve to locate the decimal point. By using scientific notation one can get rid of such zeros. For example, 0.034 gram has only two significant figures.
- Zeros sandwiched between other digits are significant, so 20.03 °C has four significant figures.
- Zeros at the end of a measurement are considered not to be significant, that is 4 500 metres has only two significant figures. If the experimenter needs to indicate that they are actually significant then he/she should use scientific notation - see the next remark.
- Zeros after the decimal point are considered to be significant. That is 12.0 A has three significant figures and 4.500×10^3 m has four significant figures.

Significant figures and measurements

In the example above we see that measuring the object with the spring scale gave us a mass of 34.4 grams while the electronic scale gave us a mass of 34.420 grams. The price of the spring scale is about R50 and that of the electronic scale R5000. So, the extra two digits in the second measurement came at a substantial price. In the experimental world it usually takes much more than just money to get a more accurate value for a certain measurement. It usually involves more time, effort, and ingenuity to get those extra digits. **Take care in reporting measurements to the correct number of significant figures.**

Calculations with measurements

Let's assume that we are actually interested in the weight, W , of the object. It can be calculated using the value 9.8 m/s^2 for gravitational acceleration. Since the second measurement came at a higher price one would expect that the weight calculated from its measurement should have more significant figures to it. That is for the spring scale we would expect that the weight should be reported as 0.337 N and for the electronic scale 0.33732 N, while keeping the answers to the same number of significant figures as the measurements. In actuality, the gravitational acceleration is also a measurement with its own uncertainty. It follows logically that the uncertainty in its value will also have an influence on the number of digits in the calculated weight of the object. So how does one decide how many significant figures there should be in a value calculated using measurements?

Significant figures in calculated values

We use some simple rules to decide how many significant figures the final value should be rounded off to. In **calculations involving multiplication, division, trigonometric and other functions** we round the final value off to the same number of significant figures as the measurement with the least number of significant figures. The weight of the object would therefore be reported as 0.34 N in both cases due to the fact that the gravitational acceleration has only two significant figures. The situation can be improved upon if we can obtain a value for g with more significant figures (more money, time and effort). As a further example if we were to calculate the value of $\sin(kx)$ using the values $k = 0.097 \text{ m}^{-1}$ and $x = 4.73 \text{ m}$ we obtain the answer 0.44 since k has only two significant figures.

When the value being calculated is obtained from **additions and/or subtractions** of values the final value is rounded off to the same decimal places as the measurement with the least number of decimal places, for example:

5.67 J	2 decimal places
1.1J	1 decimal place
0.9378 J	4 decimal places
7.7 J	1 decimal place

Exact values

Numerical constants have no uncertainty in their value. They are, therefore, not measurements and do not influence the uncertainty in a value calculated from them. Some examples are the number of millimeters in an inch (2.54 mm), which is a defined quantity (the inch) or π , which can be calculated to any required precision or the number 2 (a counting value) in the formula $2\pi r$. Such values are not taken into consideration when deciding how many significant figures there should be in the final answer.

Intermediate calculations

When a calculated value is the result of several steps of calculations one should keep (at least) one more significant digit in the intermediate answers than that which is required in the final value.

Scientific notation

Very large or small values are best reported using scientific notation. In this notation the decimal point is repositioned just after the first significant digit in the value. For example, to report 14297 J in scientific notation the decimal point needs to be positioned just after the digit 1. That is we need to divide the value by ten thousand or multiply it by 10^{-4} . In order not to change the value we therefore also need to multiply it by ten thousand or 10^4 :

$$\begin{aligned}14297 \text{ J} \\&= 14297 \times 10^{-4} \times 10^{+4} \text{ J} \\&= 1.4297 \times 10^4 \text{ J}\end{aligned}$$

Graph Rules

Rules for drawing graphs

1. The graph should have a concise title. All symbols that are used in the graph must be mentioned in words in the heading and the subject or apparatus of the experiment should be mentioned, e.g. *The temperature change, ΔT , of a sample of water as heat, Q_w , is added to it.*
2. The independent variable (the quantity that is changed **directly** by the experimenter) is plotted on the horizontal axis (the abscissa) and the dependent variable (the quantity that changes in the experiment as a result of changing the independent variable) is plotted on the vertical axis (the ordinate).
3. Clearly indicate along each axis the variable plotted on the axis in words as well as the symbol and relevant units, e.g. **Resistance, R (Ω).**
4. The scales of the two axes, not necessarily the same, should be chosen so as to allow the graph to occupy most of the available space. **The axes do not need to start at zero.**
5. The intervals should be chosen to **facilitate** plotting and reading of points.
6. Mark each experimental point with a dot with a circle around it.
7. If the points lie more or less on a straight line use a ruler to draw a straight line in such a position that the points are evenly spaced about it. This is called the **line of best fit**. Otherwise draw a smooth curve through the points. This reduces the effect of experimental errors. **Never** connect the points by means of short straight lines.

8. The gradient of a graph

The gradient (or slope) of a graph is **always** calculated by means of the definition:

$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

This definition is also used when the graph passes through the origin. **Never** calculate the slope using $m = y/x$.

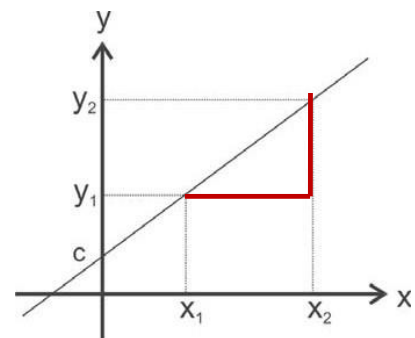


Fig.1 a Linear relationship.

- Always determine the slope from the fitted line, **never from the experimental points**.
- Indicate the slope using a **gradient triangle**.
- Choose the largest possible Δx and Δy , this allows for more accurate results. In other words, choose two points that are as far apart as possible.
- If the slope of any region of a graph is not constant, it is called a curve, as in **figure 2**. The slope of a curve at any given point is equal to the slope of the tangent at that point. The slope of the tangent is calculated in the same manner as for a straight line. The gradient of a curve is therefore not constant and depends on the independent variable.

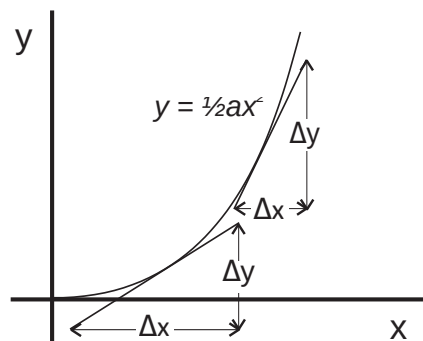
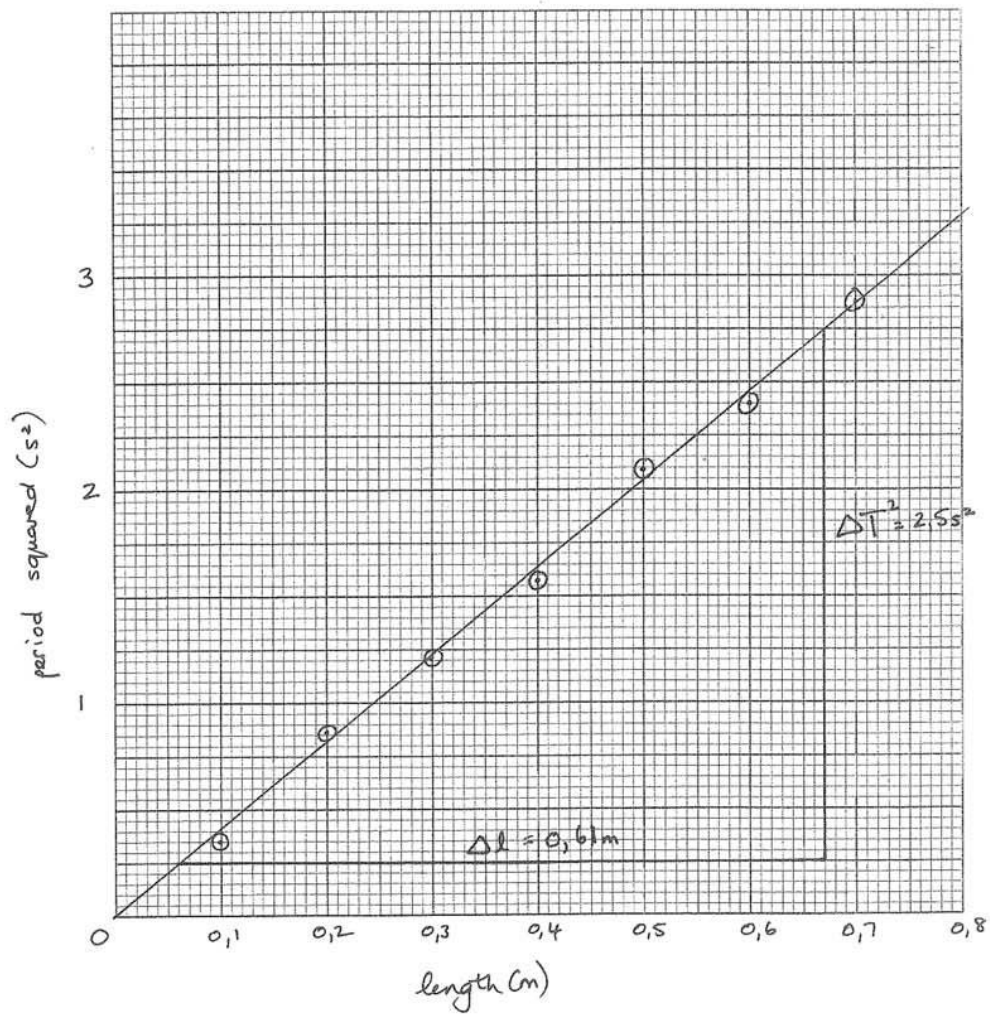


Fig. 2 The gradient of a curve.

9. If the relationship between the dependent and the independent variable is linear as in **figure 1**, then the relationship between the two quantities is of the form:

$$y = mx + c$$

The square of the period of a simple pendulum as a function of its length.



where m is the gradient of the graph and c is the y-intercept. This equation with the aforementioned constants is called the empirical (obtained through experiment procedures) equation. By comparing the empirical equation with its theoretical counterpart, it is possible to determine some useful constants.

The empirical equation of the graph in the example on page 10 is

$$T^2 = 4,1(s^2/m) l + 0$$

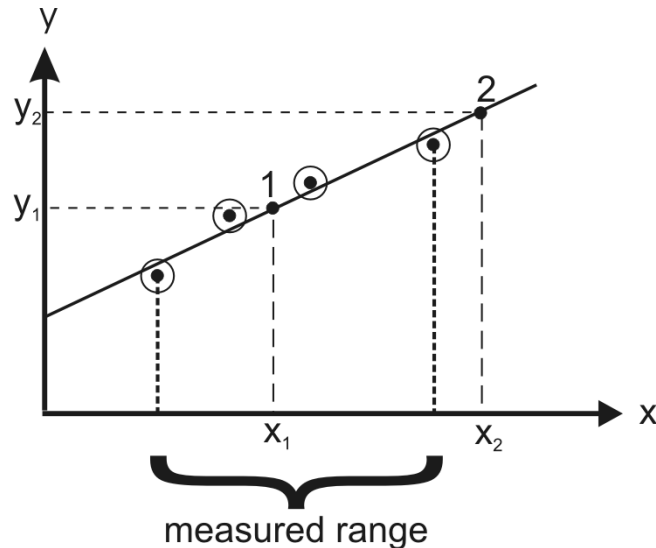


Fig.3 Interpolation and extrapolation.

1. A graph can be used to read off values for the dependent variable at values of the independent variable at which no measurement were made, like at x_1 and x_2 in **figure 3**.

If the reading is made at a value for the independent variable that is within the experimental measured range then the process is called **interpolation**, as at point 1. Interpolation leads to relatively reliable values for the dependent variable.

If the reading is made at an independent value outside of the measured range, then the process is called **extrapolation**, as at point 2. Such values are not as reliable as those obtained by interpolation since the functional relationship may deviate from a linear relationship outside the measured range. Rather extend the measuring range to include the value at which a reading is required if this is at all possible.

References:

1. Fishbane: Physics for Scientists and Engineers
2. Giancoli: Physics
3. Halliday: Fundamentals of Physics
4. Young et al: University Physics
5. Halliday et al: Fundamentals of Physics



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GRA 1: Graphs

Introduction

The quantitative information collected during an experiment is known as data. The data is used to determine if a functional relationship between the variables in an experiment exists or not. This functional relationship can be represented in at least three ways (see section 1.2, Chapter 1 of your textbook:

- a. a table of numbers
- b. an equation
- c. a graphical representation

The following section describes the important rules and techniques for representing data graphically.

The use of graphs

1. A graph is a visual representation of the functional relationship between two or more variables. We will limit ourselves to two variables.
2. Graphs may enable one to determine the mathematical relationship between the variables. This mathematical relationship is called an **empirical relationship**. If the theoretical relationship between the variables is also known then some relevant constants can be determined by comparing the theoretical and empirical equations - for example in **graph C** you will determine the decay constant, λ of some decaying process.
3. Graphs can show systematic deviations of the data from the general trend of the graph.
4. Graphs make it possible to see where small changes in the independent variable cause large changes in the dependent variable.
5. Interpolations and extrapolations can easily be made on linear graphs - for example in **graph A** in this practical, you will determine the resistance of the wire at temperatures other than those in the data.

Rules for drawing graphs

Rules for the drawing of graphs are summarised in the introductory section of this lab manual. These rules should always be applied to all graphs that you draw.

1. Straight line graphs

If a variable y is directly proportional to another variable x , i.e., $y \propto x$ or $y = mx$, the graph is a straight line which always passes through the origin (see figure 1). In the above equation, m is called the proportionality constant, and it is the gradient of the graph.

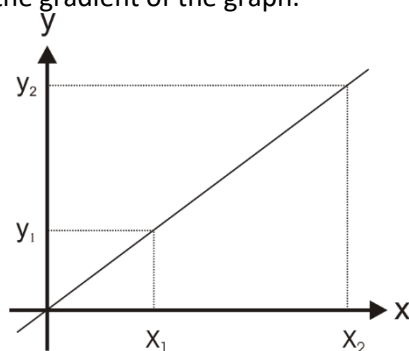


Figure 1: A directly proportional relationship.

The gradient or slope of the graph is given by:

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Remember to calculate both Δy and Δx according to the scales and the units of the relevant axes.

If the change in y is directly proportional to the change in x , then: $\Delta y \propto \Delta x$ as in figure 1.

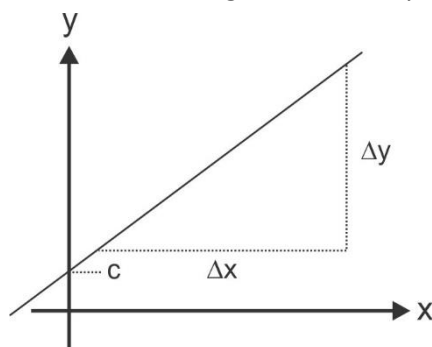


Figure 2: A linear graph with a positive gradient

If the graph is a straight line that does not pass through the origin, like in figure 2, we still say that there is a linear relationship between the variables. The graph can be represented by the equation:

$$y = mx + c$$

where c is called the intercept of the graph on the vertical, y , or dependent axis.

The gradient m is, as previously

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Graph A

The resistance R of a homogeneous wire has been measured at different temperatures, T . The results are shown in Table 1.

Table 1 The resistance of a wire at various temperatures

Temperature, T (°C)	Resistance, R (Ω)
-20.0	307
10.0	325
30.0	338
45.0	353
65.0	362
75.0	375
90.0	382

1. Draw a graph of the resistance, R , of the wire as a function of its temperature, T , that is $R(T)$. The temperature axis should include zero °C, but for the sake of convenience, the start of the resistance scale should not be zero. Insert your graph on the next page.

Now that you have drawn a graph of the data and found that there is a linear relationship between the variables, you can formulate the mathematical relationship between the variables. This mathematical formula is called the empirical relationship since it was determined using experimental data.

2a. Determine and clearly indicate on your graph, the change in resistance per unit change in the temperature, that is $\Delta R/\Delta T$, or the slope of the graph. Choose points 1 and 2 as far as possible from each other - see figure 1 - and **do not** use data points from the table. Do not forget the units of the gradient when you state its value.

2b. Set up the empirical equation for the resistance of the wire at different temperatures, i.e., the formula relating the relationship between the resistance and temperature of the wire. This equation should have values and units for both constants in the linear equation. Use the symbols R and T for the variables in the equation.

2c. Now that you have determined the relationship between the measured quantities, you can either use the graph or the empirical equation to determine values for the resistance of the wire at other temperatures. Let's use the graph to determine the resistance of the wire at some temperature. You should indicate **on the graph** where the readings were taken:

- i. The resistance of the wire at $-30\text{ }^{\circ}\text{C}$. You are extrapolating in this case. That is, you are assuming that the functional relationship between the variables will remain linear outside of the measured range. Such an assumption is risky since you do not know whether the relationship is still linear outside of the measured range of values.
- ii. The resistance of the wire at $25\text{ }^{\circ}\text{C}$. Now you are interpolating, or predicting that the resistance will take on this value, a much safer bet since this is within the measured range of values.

2. The power function: $y = kx^n$

The power function is often found in physics, for example:

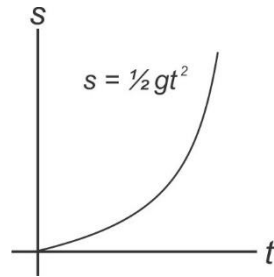


Figure 3: Freefall out of

- a. The distance, s , that an object, initially at rest, moves under the influence of gravity during a time interval t is given by:

$$s = \frac{1}{2} gt^2$$

- b. The period, T , of a simple pendulum depends on its length, L , according to:

$$T = k\sqrt{L}$$

- c. The force, F , between two point charges or masses separated by a distance, r , is given by an inverse square law:

$$F = k \frac{1}{r^2}$$

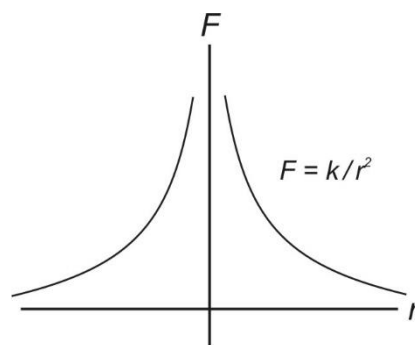


Figure 4: The electric force between two point charges

Advantages of linear representations

- a. It is easy to fit data points to a straight line.
- b. The formula which describes the relationship between the variables can be more easily established.
- c. Interpolation and extrapolation - determining values for the dependant variable other than in the measurement set - can easily be made on a straight-line graph compared to on a curve.

It is therefore desirable be able to transform a power relationship graph into a straight line, which will be discussed next.

Transforming power relationships to straight line graphs

Starting with the general power relationship:

$$u = kv^n$$

Taking the logarithm on both sides:

$$\log u = \log(kv^n)$$

$$\log u = \log k + n \log v \dots (1)$$

$$Y = C + MX$$

We obtain a linear relationship between the logarithms of the dependent, u , and independent values, v , that is, we get a linear graph from the “new” set of data values, $X = \log(v)$ and $Y = \log(u)$. From such a linear graph, we can then determine the coefficient, k , and the exponent, n , of the original power relationship. That is the y-intercept of the graph is: $C = \log(k)$ and the gradient of the graph $M = n$.

Alternatively, if there is a power relationship between the original measured quantities, u and v , then a graph of u against v^n will result in a straight-line graph with a slope of value k , if the correct value of n is used.

Graph B - Fitting a power function to data

We will investigate the relationship between the semi-major axis, a , of the orbits of some of the moons of Saturn and the time, or period, T , to complete their orbits around the planet in earth days (24 hrs).

Let's use the data of the moons in Table 2 to investigate whether there is possibly a power relationship between the given variables.

Table 2: The semi-major axes and periods of orbit of some of Saturn's moons.¹

Moon	a (x10 ³ km)	T (days)
Rhea	527	4.52
Titan	1 222	16.0
Hyperion	1 481	21.3
Iapetus	3 561	79.3
Phoebe	12 952	550

You are going to try and fit the data to a power function of the form

$$a = kT^n \dots\dots\dots(2)$$

$$\therefore \log(a) = \log(k) + n \log(T)$$

to the data where k and n are constant values which you need to determine.

3. Plot the data on the given log-log graph paper.

According to your graph, can you fit a power function to the data? Note that a graph of $\log(a)$ vs $\log(T)$ drawn on normal graph paper should then also be linear.

4a. Determine the gradient of the graph according to

Gradient = n

$$= \frac{\log(a_2) - \log(a_1)}{\log(T_2) - \log(T_1)}$$

$$= \frac{\log(a_2/a_1)}{\log(T_2/T_1)}$$

4b. Read the y-intercept of the graph off where the period has the value $T = 1.0$ day (why?). This is the constant k in the equation above. Why don't we take the log in this case? The units of k can be determined by considering the units of the variables, a and T , assuming the relationship given in **equation (2)**.

4c. Next, set up the empirical relationship between the distance and period values. Supply values for the constants, together with their units, in the equation.

4d. What can you conclude about the mathematical relationship between a and T , according to your graphical analysis?

¹ Source: https://www.esa.int/Science_Exploration/Space_Science/Cassini-Huygens/Saturn_s_moons



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GRA 2: Graphs

The exponential function $y = a^x$

Exponential functions are described by a mathematical relationship of the form $y = a^x$. Note that the variable is in the exponent. a is the base and it is raised to the power of the controlling variable, x . We use only positive values for the base.

We are dealing with an exponential function whenever the growth rate of a certain quantity is constant. For instance the growth rate of the human population is approximately 1,53% per year. That means that if the population was p_1 in a particular year it would be $1,0153 p_1$ in the following year. The startling nature of this type of function is that although the growth rate might seem small the quantity itself increases (or decreases) quite fast (exponentially) over time. Examples of such processes are:

- The decay of radioactive material. The amount of radioactive isotopes decreases exponentially with time as they mutate to other elements. In this case the relation is of the form $y = a^{-x}$ to reflect that the quantity is decreasing.
- The growth rate of a colony of bacteria, given enough resources, is directly proportional to the size of the colony at that particular time. That is, the amount by which they change per unit time, or db/dt , becomes more as the amount of bacteria that is present increases, or $db/dt \propto b$. This is evident in the fact that the gradient of the graph of the amount of bacteria present as a function of time increases as the amount of bacteria increases - see figure 1. The proportionality constant is the growth rate, r , of the colony (which is constant). We incorporate the growth rate as follows into the exponential function

$$b = a^{rt}$$

In this way, it controls how fast the function is increasing.

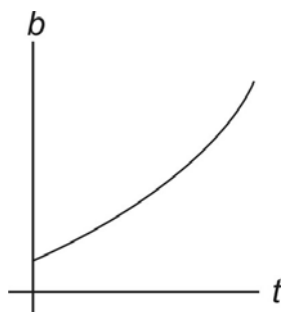


Figure 1: The amount of bacteria in a colony with time

- The emptying of water from a tank.
- The reduction in air pressure with height.
- The discharging of a capacitor in a circuit. The rate at which the capacitor discharges decreases as the amount of electric charge on it decreases, similar to the rate at which a colony of bacteria changes.

For naturally occurring processes we often find that the base $e = 2,71828$ should be used. The exponential function then becomes

$$y = y_0 e^{-rx}$$

for decreasing functions and

$$y = y_0 e^{rx}$$

for increasing functions. y_0 is the starting value for the function, that is when the control variable, x , is zero.

Logarithms

If $y = a^x$ then we say that the logarithm of y to the base a is x or $\log(y) = x$. The logarithm of a number is therefore the exponent to which the base must be raised in order to obtain that number.

Common logarithms are to the base 10. In such a case the base is not indicated explicitly, for example $\log(100) = 2$. We speak of natural or Napierian logarithms when the base is e . In such a case we write $\ln$ rather than $\log$, for example $\ln(2,00) = 0,693$.

Transforming graphs of exponential functions to straight-line graphs

If one physical quantity, y , is decreasing exponentially with respect to another, x , then

$$y = y_0 e^{-Zx}$$

Taking the natural logarithms on both sides of this relationship we get

$$\begin{aligned} \ln(y) &= \ln(y_0) + \ln(e^{-Zx}) \\ \ln(y) &= \ln(y_0) + (-Z)x \dots \dots \dots (1) \end{aligned}$$

$\begin{array}{ccccccc} \uparrow & & \uparrow & & \uparrow & \uparrow & \\ y & = & c & & + & m & x \end{array}$

We can therefore transform an exponential graph to a linear graph if we plot a graph of $\ln(y)$ as a function of x . Note that the gradient of the linear graph will be the negative of the decay constant, λ , that is $m = -\lambda$, and its y-intercept will be the natural logarithm of the starting value or $c = \ln(y_0)$.

Semi-logarithmic graph paper

To save the effort of first calculating the logarithms, special graph paper called semi-logarithmic graph paper has been designed with the grid lines drawn at logarithmic intervals along one axis and linear divisions along the other. Such graph paper is used to draw exponential functions.

Graph C - Fitting an exponential function to data

Quantities that increase or decrease exponentially are relatively common. The question is, how would one recognize it from a set of data and how would one set up the mathematical relationship between the variables? In this section, we intend to answer these questions.

The count rate of a radioactive sample was measured with a radiation detector over a period of seven hours. The data is displayed in **Table 1**. We intend to fit the data to the exponential function:

$$R = R_0 e^{-\lambda t} \dots\dots\dots(1)$$

where R_0 and λ are constants.

Table 3: The decreasing activity of a radioactive sample.

Time, t (hours)	Count Rate, R (counts/s)
1.0	4803
2.0	3010
3.0	1890
4.0	1233
5.0	754
6.0	472
7.0	319

1. Draw a graph of the count rate, R , of the sample as a function of time, t . Try and draw a smooth curve through the data points. How can you tell if the relationship is linear or not?

Make a new table with x-values, the time, in hours and the y-values, the natural logarithm (that is the logarithm to the base e) of the count rate values.

2. Draw a graph of this new set of values (insert it on the next page). It should be linear if the original data can be represented by an exponential function. Note that one would also obtain a linear function if the data were to be plotted on a semi-logarithmic graph paper. The fact that the second graph is linear confirms our suspicion that you can fit an exponential function to the data.

3a. Explain how you know that there is a linear relationship between the natural logarithm $\ln(R)$ and time, t , values. Determine the y-intercept, y_0 , and the slope, α , of the graph. You need to choose arbitrary points 1 and 2 (not data points!) for the slope as far apart as possible on the graph. Show your full calculations and values.

3b. Write down the linear relationship between the $\ln(R)$ and t values and substitute the constant values you found for the y-intercept and slope into the linear equation. Display the units, if any, of the inserted constants.

4a. Show that

$$\ln(R) = \ln(R_0) + (-\lambda)t \dots\dots\dots(2)$$

by taking the natural logarithms on both sides of equation 1. Compare this equation to your empirical equation in question 3b. Explain why $y_0 = \ln(R_0)$ and $\alpha = -\lambda$. Use this information to determine the corresponding values, and units, for R_0 and λ .

4b. Write equation 1 down again with the corresponding values, and units, for the constants in the equation. This is the exponential function for the data.

5a. At this point you can say that the radioactive sample is decaying exponentially according to equation 1. Explain what the meaning of the half-life, $T_{1/2}$, of this process is.

5b. Next use equation 1 to show that the relationship between the half-life and the decay constant, λ , is

$$\ln(2) = \lambda T_{1/2} \dots\dots\dots(3)$$

5c. Calculate the value of the half-life, $T_{1/2}$, using equation 3. Include the units for the half-life.

Summary

For an exponential relationship between the variables, we obtain a linear graph on normal graph paper if we plot the natural logarithm of the dependent values versus the independent values. An exponential function also produces a straight-line graph if it is drawn on semi-logarithmic graph paper.

Power functions are linear if drawn on log-log graph paper. On normal graph paper they produce a linear graph if one plots the common logarithm of the dependent values versus the common logarithm of the independent values.

Given a set of measurements of two quantities, how would one determine the empirical relationship between them by drawing a graph?

- First plot the data on axes with linear divisions. One can readily set up the empirical relationship if such a graph is linear. It is a direct proportion.
 - If a straight line is not obtained, plot one set of values against the inverse of the other. If this is a straight line, we have an inverse proportion.
 - If a straight line is still not obtained, we resort to graphs in which the divisions on one or both axes are logarithmic.
 - First try a semi-log graph: One axis has linear divisions, and the other axes has logarithmically spaced divisions. A straight line suggests that one can fit an exponential relationship to the data.
 - If this is unsuccessful one can plot the data on a double - log plot, where both axes have divisions that are logarithmically interspaced. A straight-line graph then suggests a power relationship can be fitted to the data.
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DEPARTMENT OF PHYSICS

FSK 116

Practical Report

MET – Measurements and Units

Introduction

Please refer to the *Lab Introductory Notes* in this document for the rules you need to follow in this practical. Also read the “*Vernier Caliper Notes*” at the end of this document to learn how to read the Vernier calipers. **Note** that your report should follow the experimental report format outlined in the Lab Introductory Notes.

Study Aims

You must be able to:

1. Use the vernier caliper with the necessary dexterity and accuracy to make measurements.
2. Convert measurements to S.I. units.
3. Round calculated values off to the correct number of significant figures.
4. Use the experimental report format outlined on page 2 of the lab manual.

1. EXPERIMENTAL AIM:

RESULTS

2. Significant figures

Indicate the number of significant figures in the following values:

- 2.1. 7.2×10^3 km
- 2.2. 0.07800 m/s
- 2.3. 900 kg
- 2.4. What is the difference between the following two measurements: 7 km and 7.0 km?

Use the densities in Table 1 for the following calculations. The values in the table are also measurements so their significant figures need to be taken into account in your final answer. Work in S.I. units and round your final answers off to the correct number of significant figures.

- 2.5. Calculate the mass of a bakelite plate with dimensions 43.0 cm x 62 mm x 1.000 m
- 2.6. Calculate the mass of a spherical perspex bead with a diameter of 17.0 mm.

Table 4 The densities of various material

Substance	Density ($\times 10^3 \text{ kg/m}^3$)
air	0.00120
wood	0.3 to 0.9
ice	0.92
polyethylene	0.880 to 0.940
water	1.00
blood	1.06
perspex	1.18
bakelite	1.33
PVC	1.3 to 1.45
bone	1.7 to 2.0
teflon	2.20
aluminium	2.70
glass	2.5 to 4.2
iron	7.87
stainless steel	7.75 to 8.1
brass	8.40 to 8.73
copper	8.96
silver	10.5
lead	11.34
mercury	13.6
uranium	19.1

3. Vernier caliper readings

3.1. Assume that the pointer in figure 1 slid along the ruler from the 0 mark up to the indicated position. How far did it move along the ruler? Include an estimate for the fraction of the millimeter distance, Δ , in your value for d . Give your answer in millimeters.

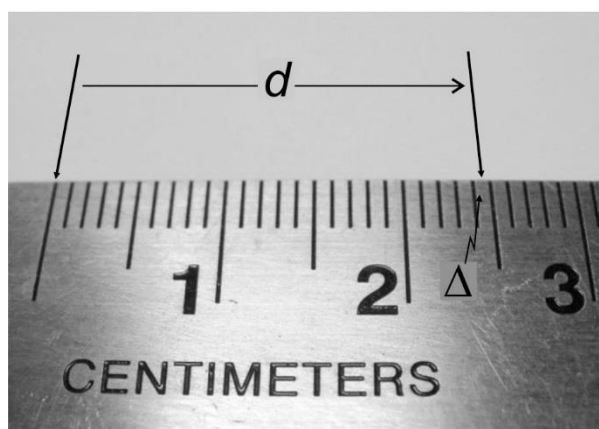


Figure 1 The distance a pointer moved through

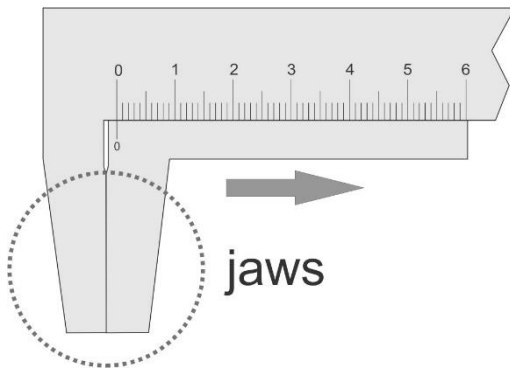


Figure 2 Vernier caliper with jaws closed

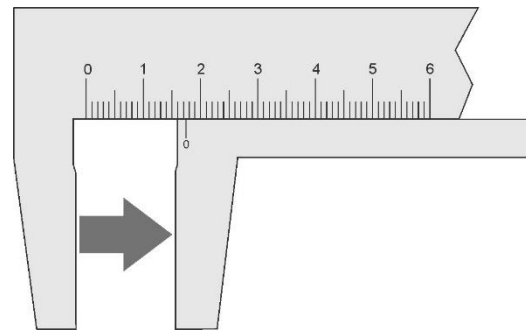


Figure 3 The Vernier caliper with its jaws opened

- 3.2. The jaws of the caliper are closed in **figure 2**. The bottom jaw slides along the main cm scale just above it – see **figure 3**. The 0 mark on the bottom jaw and the 0 mark on the cm scale are aligned with each other when the jaws are closed – **figure 2**. In **figure 3** the jaws were opened to the indicated position. How many millimeters were they opened, that is how far did the 0 on the bottom jaw slide along the cm scale above it? Estimate the fraction of the millimeter value, Δ , again.

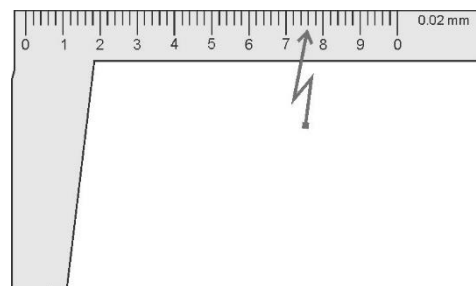


Figure 4 The movable jaw of the caliper with the engraved vernier scale indicated on it.

- 3.3. **Figure 4** shows the movable jaw of the vernier caliper again. On it is an engraved scale where each small division represents 0.02 mm. We say that the least count or resolution of the caliper is 0.02 mm. The line to the right of the 0 line is thus the 0.02 mm line. To its right is the 0.04 mm line, then the 0.06 mm followed by the 0.08 and the 0.10 mm with a 1 digit indicated next to it. What vernier line is the arrow pointing to on the scale in **figure 4**?

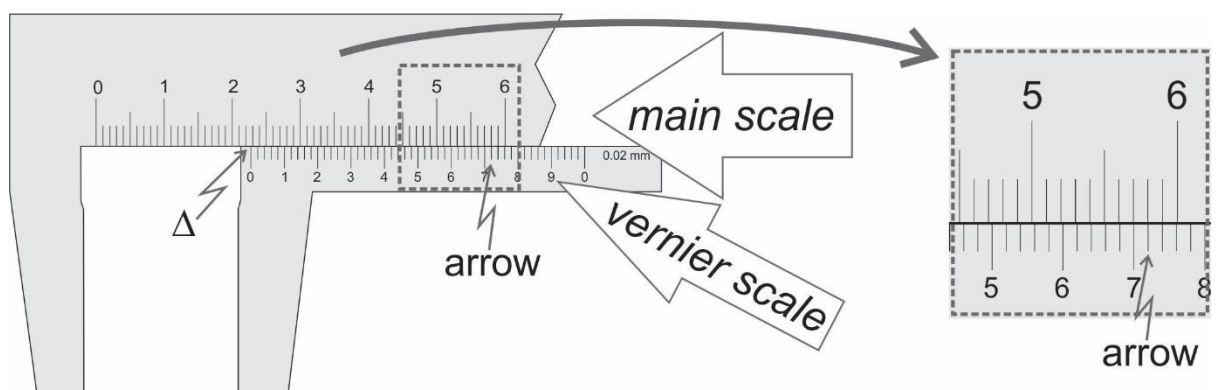


Figure 5 Reading the vernier scale of the caliper

3.4. The vernier scale enables one to accurately determine the value of, Δ , the amount by which the 0 on the vernier scale went beyond the millimeter mark on the top, main or cm scale – the Δ bit in **figure 5**. The accurate value of Δ is read off on the vernier scale and added on the reading made on the main scale. In order to determine the value of Δ we compare the position of the lines on the vernier scale to the position of the lines on the main scale - **figure 5**. Start at the left-hand end of the vernier lines and move your eye along them to the right – **figure 5**. Notice that at some point the two sets of lines come into alignment with one another - at the arrow's position. The vernier scale is read off at this alignment point, the position of the arrow. What is the vernier scale reading at this alignment point? Add this fraction of a millimeter value to your reading on the main scale - where the 0 of the vernier scale is located on the main scale - and note the final value down in your book. Note that a reading made with such a vernier caliper must always have a figure for the 1/100th of a millimeter digit! Even if the reading falls exactly on a millimeter mark, in which case you will have a 0 in the tenth and a 0 in the hundredth of a millimeter position in the reading.

3.5. What is the vernier caliper reading in **figure 6**?

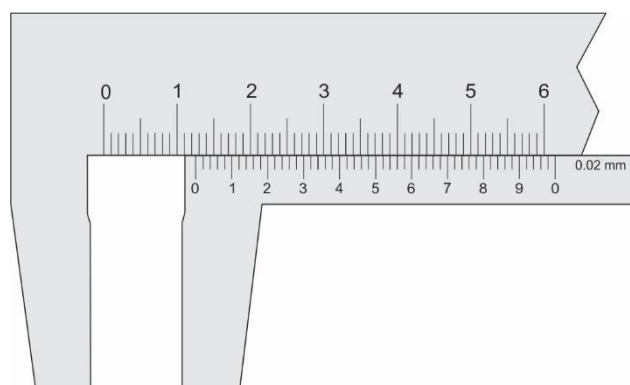


Figure 6 Another reading on a vernier caliper

3.6. The least count or resolution of the vernier caliper in **figure 7** is 0.05 mm, not like the previous examples that were 0.02mm. In this case the vernier divisions represent 0.05 mm. What is the reading in this case?

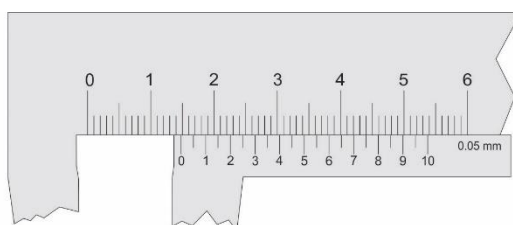


Figure 7 A vernier caliper with a 0,05 mm resolution

Below are two links to a virtual simulator for the vernier calipers. You can try the simulation yourself and you can confirm your readings

<https://iwant2study.org/ospsg/index.php/interactive-resources/physics/01-measurements/5-vernier-caliper>

https://www.stefanelli.eng.br/en/virtual-vernier-caliper-simulator-05-millimeter/#swiffycontainer_2

4. Density determination

To determine the density, ρ , of the material of a piece of metal square tubing. The density can be calculated from

$$\rho = \frac{m}{V}$$

where m is the mass of the tube and V its volume.

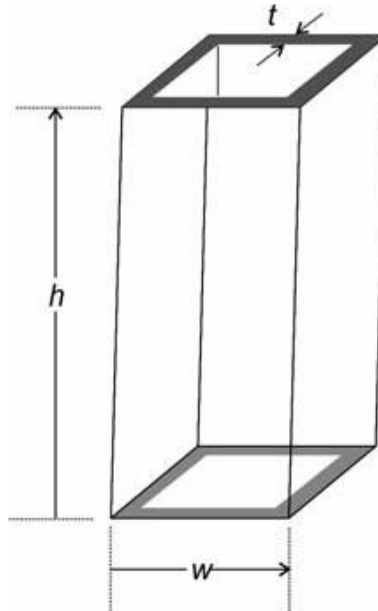


Figure 8: A piece of metal square tubing

Set up a formula for density, ρ , of the hollow tube using the indicated symbols on the figure.

Measure the mass of the hollow tube on the electronic scale and record the value. Check that the scale is level and zeroed.

- 4.1. Measure each of the dimensions of the hollow tube - **figure 8** - three times. Convert the measurements to S.I units and tabulate them. Calculate the average of each dimension. Tabulate these values and include a row for the average of each of the dimensions.
- 4.2. Calculate the density of the hollow tube. Round the value off to the correct number of significant figures and write down the units of the value.

Look in **Table 1** for materials with similar densities. What can you conclude from this?

4.3. CONCLUSION:



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Practical Report

CLM: Conservation of Linear Momentum

Introduction

In this experiment you will arrange a collision between two flat round plastic discs, one large one, S – the striker, and one smaller one, T - the target - see **figure 1**. You will use an elastic band to shoot the striker towards the stationary target. After the collision the discs will be moving off in different directions with respect to each other.

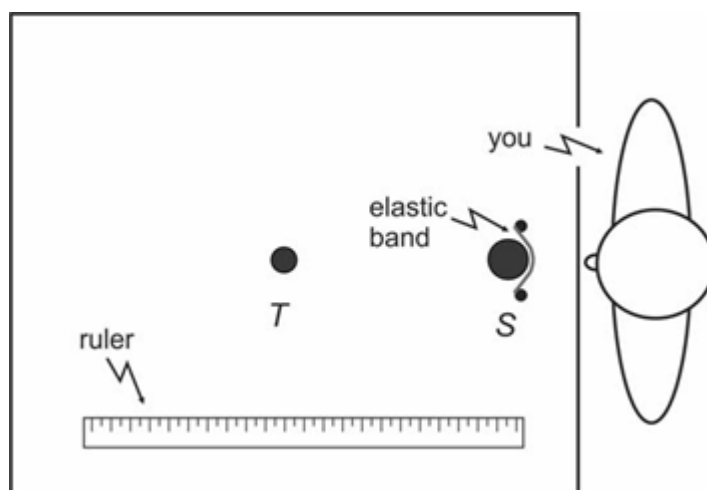


Figure 1: A view from the top of the target disc, T , and the striker disc, S , on the board before the striker is shot towards the target using the elastic band.

We will be interested in the combined or resultant momentum of the two discs, the “system”, before and after the collision. That is, is the combined, or resultant, momentum of the system before the collision the same as the combined momentum of the system after the collision? Why should they be the same? Wouldn't the collision change the resultant momentum? The momentum of a system of objects can **only** be changed by an **external force** acting on it, that is a force that originates from outside of the system. When the discs collide, they experience collision forces from each other, but these forces are an action-reaction pair so they cancel each other out **if we consider the discs as a system**. The collision forces are internal forces and cannot change the momentum of the system. This means the collision forces will not change the resultant momentum of the system of discs. We say that the momentum of the system is conserved.

Safety Precautions

Please be careful when you get up on the laboratory chair and again when you step off it. Rather support yourself with at least one hand on the bench while you get on and off. Place the leading foot in the middle of the chair and not off to the side of the chair. Best someone supports you while you climb onto and get off the lab chair. Do not use the metal arm with the lamp above the bench to support you, it is too weak.

Apparatus

- One large wooden board with a piece of paper taped on top of it. The board should have no dirt on it and the paper should be as flat and tight as possible.
- Two round plastic discs, one large and one small. The large disc is placed on the paper with the dimple, for your shooting finger, on top.
- A wooden ruler for calibration purposes.
- A protractor used to draw the vector arrows in their respective directions in this report.
- An elastic band.
- A digital camera to record the collision and a cable to download the videos onto the computer.
- A lamp mounted above the board.
- A computer with the Tracker program installed on it.

Filming the collision

1. Place the objects on top of the paper on the board as in **figure 1**.
2. Delete the videos on the camera's memory card and select the high definition, HD, movie mode. Also set the camera in the Auto mode.
3. Mount the camera above the wooden board next to the lamp on the metal arm above the bench. The camera should be pointing down towards the board.
4. Slide the board so that the target disc is positioned below the lens of the camera. Make sure that the ruler is included in the camera's image.
5. Switch the lamp on.
6. Practice shooting the striker towards the target so that both discs move off at an angle with respect to each other after the collision.
7. Make recordings of several successful collisions.

Getting ready to analyze the collision

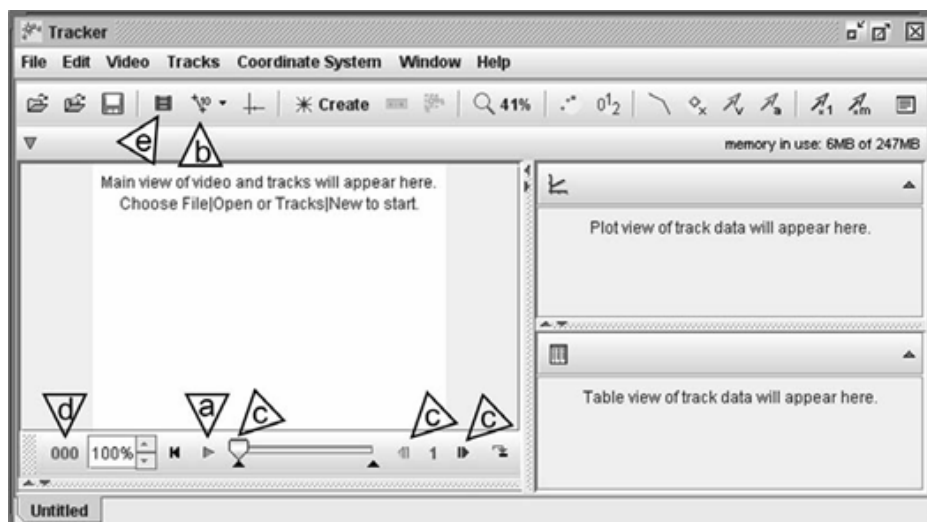


Figure 2: The Tracker program window. Some of the control buttons used in the experiment are marked - a, b, c, d and e.

1. Download the videos onto the computer and switch the camera off.
2. Run the Tracker program on the computer and open a video in it.
3. Play the video in the program, at [a] in **figure 2**. Repeat for the other videos and select the best video for subsequent analysis.
4. Create a new calibration tape in the program - at [b] in **figure 2**. Drag the tape's endpoints to the ends of the ruler in the video. Change the length of the tape to the length of the ruler (click on the value next to the tape, type in the length of the ruler and hit enter). Note that you can use the mouse wheel to zoom in or out of the video. The display can also be dragged around after you have zoomed in. The tape is now calibrated and can be used to measure distances and directions in the video.

Determining the velocities of the discs

Next you are going to use the calibration tape, a software tool in the program, to measure the speeds of the discs

$$v = \frac{\Delta s}{\Delta t}$$

where Δs is the distance a disc moved from one frame to the next frame, that is the video consists of photographs, or frames, taken in quick succession. And Δt is the elapsed time between the two frames. You will also determine the direction, θ , in which the discs were moving, see **figure 3**.

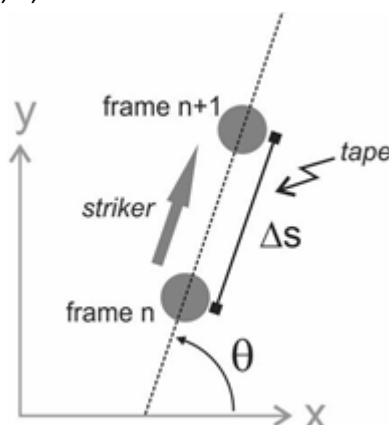


Figure 3: The position of the striker in two subsequent frames of the video.

1. Step through the frames using the three [c] controls indicated in **figure 2** and locate two frames just before the collision in which the striker is seen. Note that the frame numbers are indicated at [d].
2. Step to the first of the two chosen frames. Note that the graphics do not synchronize up immediately when you step back in the video, so step back several frames and then forward to the one you are interested in, the frame number, in order to let it synchronize again.
3. Drag one endpoint of the tape to the middle of the striker. You can zoom and drag the display to increase the accuracy if necessary.
4. Step to the next frame and drag the other endpoint of the tape to the middle of the striker. The program now displays the values of Δs , the length of the tape and θ , the direction of the tape.



Figure 4: The program displays the length and direction θ , of the tape

5. The x-axis, not shown in the display, is oriented horizontally, and pointing to the right or east and the angle is measure anticlockwise with respect to the axis. Negative angles are measured clockwise with respect to the axis. If the displayed angle does not seem to make sense, then the program is assuming that the object moved in the opposite direction. In such a case you need to turn the tape the other way round by swapping its endpoints on the images of the two striker frames.
6. The time between frames, Δt , stays constant throughout the video and can be viewed by opening the clip inspector window at [e].
7. Now repeat the measurements for both discs after the collision. Tabulate your results. Add another 3 columns to your table, one for the speed and two for momentum.

1. Experimental Aim:

RESULTS

2. Table of results

The momenta of the discs

In this part of the experiment, we represent the momenta with arrows, which you will draw in this report. Momentum is a vector quantity, and we need special mathematics to handle vectors. Here we will treat them graphically by drawing them as arrows in this report.

1. Measure the mass of the disc. Note the values down and calculate and tabulate the magnitude of the momenta in each case. The directions of the momentum vectors are the same as the directions that the discs moved in, the measured θ s.
2. The length of the momentum arrows that you are going to draw in this report represents the magnitude of the momenta. You therefore need to use a scale factor, for example 1000 kg m/s is 1 meter long, in order to calculate the length of the arrows. Choose a scale factor that will give arrow lengths that are not too long or too short, say between 5 and 10 cm. Calculate and tabulate the arrow lengths using your chosen scale factor. Also note the scale factor down in this report.
3. The resultant momentum of the system of discs before the collision, $\vec{p}_1$, is just the momentum of the striker before the collision. If momentum was conserved the resultant momentum of the discs after the collision, $\vec{p}_2$, should produce the same vector, that is both the magnitude and direction of $\vec{p}_1$ and $\vec{p}_2$ need to be the same. So, we have to graphically combine the two momentum vectors after the collision, $\vec{p}_{s2}$ and $\vec{p}_{T2}$ to produce the resultant momentum of the system after the collision $\vec{p}_2$. How to do this?
 - a) Construct the arrow that represents the **momentum of the striker after the collision**, $\vec{p}_2$, as follows. It might be easier to use graph paper at the back of your report for these constructions. First decide where the arrow's tail is going to be and mark the direction, θ_{s2} , out with a protractor. Then mark the length, L_{s2} , off along this direction - see **figure 5**. Note: Your arrows may point in other directions, but the construction will be similar – head-to-tail. Add an arrowhead at the end and write the vector symbol $\vec{p}_2$ next to it. This arrow represents the momentum of the striker after the collision.

b) Next, we are going to add the **momentum of the target after the collision**. Mark its direction, θ_{T2} , off at the head of the arrow of $\vec{p}_{S2}$. Mark the length, L_{T2} , off and complete it with an arrowhead and $\vec{p}_{T2}$ symbol.

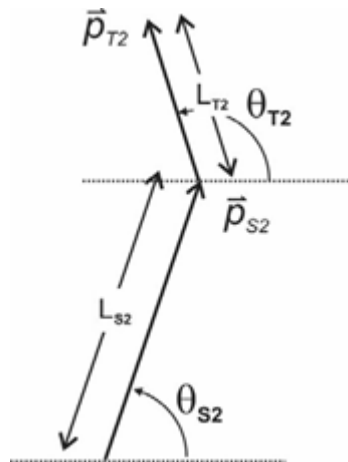


Figure 5: Constructing the momentum arrows representing the momenta of the discs after the collision.

c) Next, we want to construct the **resultant momentum of the system of discs after the collision**, $\vec{p}_2$. Draw another arrow on your vector diagram starting at the tail end of $\vec{p}_{S2}$ and end its head at the head of $\vec{p}_{T2}$ - **figure 6**. Label it as $\vec{p}_2$, since it represents the resultant momentum of the discs after the collision. Note that if we started our constructions with $\vec{p}_{T2}$ and then joined up with $\vec{p}_{S2}$ we would have obtained the same vector arrow - **figure 6**. This will give us just the other half of the parallelogram.

4. Now we want to compare the momentum of the system before the collision $\vec{p}_1$, to the momentum of the system after the collision, $\vec{p}_2$, in order to see if momentum was conserved. Draw your $\vec{p}_1$ arrow next to your vector diagram by starting out with its tail at the tail end of $\vec{p}_2$, that way $\vec{p}_1$ and $\vec{p}_2$ form a V shape.

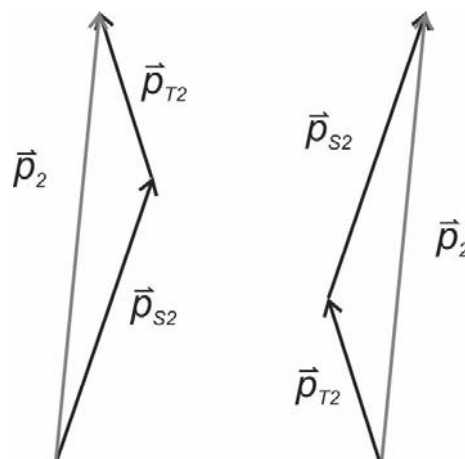


Figure6: Constructing the resultant momentum arrow of the discs after the collision, $\vec{p}_2$.

3. Vector diagrams

DISCUSSION

4. Answer the following questions:
 - a. Do the discs travel with a constant speed along their tracks?
 - b. Do you therefore expect that the resultant momentum of the system of discs will be conserved? Why?
 - c. Name the external force that will alter the resultant momentum of the system of discs.
 - d. Explain why we should measure the speed of the discs just before and just after the collision and not when they are far away from the point where they collided.

CONCLUSION

5. Are the $\vec{p}_1$ and $\vec{p}_2$ vectors identical or equal to each other, or in other words was momentum conserved? Explain your answer.

Physicists have so much confidence in the principle of conservation of momentum that they use it to postulate the existence of particles that went undetected in collisions arranged in particle accelerators. They know that such particles must exist in order to account for the “missing” momentum.

More information

See the Large Hadron Collider (LHC) webpage <https://home.cern/topics/large-hadron-collider> for more information on the world’s largest particle accelerator where high speed collisions of atomic particles are produced and analyzed.

Tracker

Tracker is a very powerful video analysis and modelling program that can be downloaded from <http://physlets.org/tracker/>

There are also extensive help pages on the use of the program. An instructional video on the use of Tracker in this experiment should also be available in the Practicals folder on your Physics clickUP web page.

The camera

The camera used in this experiment is either the Canon Powershot A810 or A2300. The Getting Started guide (the User guide gives more information on these topics) can be downloaded from <https://www.canon-europe.com/support/business-product-support/>

It is a good idea to delete all pictures and videos on the camera before you start with the experiment. This can be done by formatting the memory card.



DEPARTMENT OF PHYSICS

FSK 176

Practical Report

ELA: Elasticity

Study aims

You must be able to:

1. Investigate the elasticity of a steel helix spring and a rubber band.
2. Draw a graph to demonstrate the relationship between the applied load force and the extension of the spring.
3. Use Hooke's law to find the force constant, k , of the spring.

Introduction

The elasticity of a body is the ability of the body to return to its original size and shape after being distorted by some force. If, after distortion, the body recovers its size and shape completely, it is said to be perfectly elastic. From everyday observations we know that some objects, like helical springs and rubber bands stretch when a force is applied to them. When the force is removed, the objects appear to return to their original shape.

Hooke's law

When a spring is stretched in the y -direction, it exerts a **restoring force** $\vec{F}$ (elastic force) which is directly proportional to the extension $\vec{y}$. The relationship between the restoring force and the extension is given by **Hooke's law**:

$$\vec{F} = -k\vec{y} \dots\dots\dots (1)$$

k is the **force constant**. The minus sign indicates that the restoring force **opposes** the extension. If we are only interested in the magnitudes of the quantities, then we can simplify this relationship to:

$$F = k y \dots\dots\dots (2)$$

At equilibrium we find that the load force W is the same as the restoring force F , therefore:

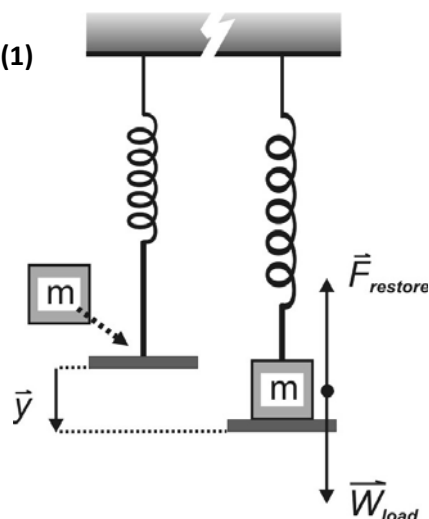


Fig. 1 The extension of a spring by a load force.

$$W = k y \dots\dots\dots(3)$$

Experiment

1. Experimental Aim:

A. Elasticity of a helix spring

- ❖ Suspend the mass hanger, mass m_0 , from the spring.
- ❖ Take a reading, y_0 , on the ruler by using the bottom of the hanger as a reference – **figure 2**.

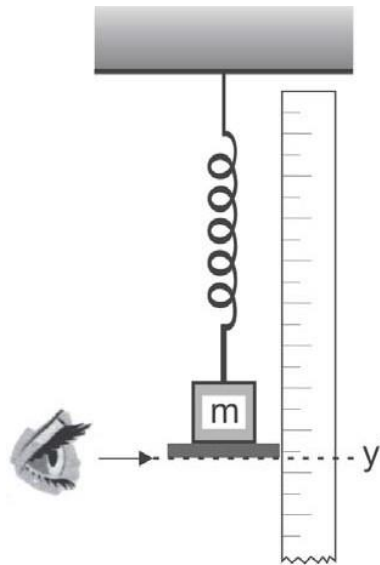


Fig.2 Measuring the position of the extension of the spring.

2. Table 1

- ❖ Load the spring in steps with 50 g masses and note the corresponding readings, y' , on the ruler. Do not exceed a total load of 250 g (*excluding the mass of the hanger*). Take readings of the position of the spring, y' , and tabulate them.
- ❖ Remove the masses from the hanger one by one and again take readings of the position of the spring, y' , on the return path (*do not tabulate*). Does the spring return to its original length?
- ❖ Calculate the values for the **extension y** ($y = y' - y_0$) against and tabulate them.
- ❖ Tabulate the values for the **load m** , where m is the total mass placed on the hanger. (*Note that m does not include the mass of the hanger*).
- ❖ Calculate the load forces, W , exerted by the masses on the spring from the relationship $W = mg$. Take $g = 9,8 \text{ ms}^{-2}$. Include these in the table too.

Remark: Which of the two variables, y and W , is the **independent variable**? The independent variable is the variable displayed on the **x-axis** of the graph, while the dependent variable is on the y-axis. The independent variable is the one that **is changed directly by you** the experimenter, and the dependent variable then changes as a result of the change you made to the setup.

3. Graph

Draw a graph of the extension, y , and load force, W , values. Put the dependent variable on the y-axis and the independent variable on the x-axis. Draw a best fit line through the data points. (*Remember to follow all graph rules*).

Does the spring obey Hooke's law?

Remark: Indicate the axis subdivision with tick marks like in **figure 3**. Also the scale divisions **should be only one of these distances apart** 1.0, 2.0, 5.0, 10.0 or powers of 10 of these like 0.1, 0.2, 0.5 or 10, 20, 50, 100 and **no other distances at all**. This will make it easier to plot and take readings on the graph. For instance, if you choose to make the scale divisions 0.2 apart then the values at the ticks will be 0, 0.2, 0.4, 0.6 ..., that is multiples of 0.2. How does one decide which scale divisions to use on the graph? You should choose one of the suggested scale divisions which will result in the data points covering as large a portion of the graph paper as possible.

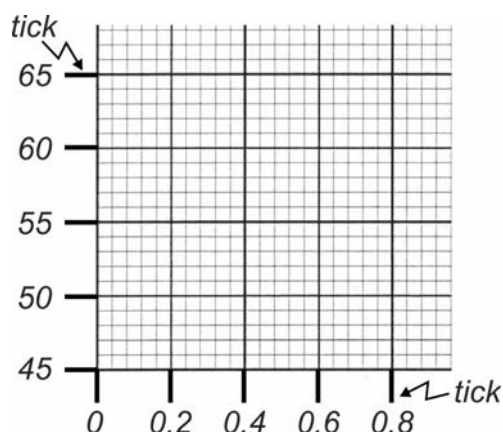


Fig. 3 A graph with 0.2 units scale divisions for the independent variable and 5 units scale divisions for the dependent variable.

4. Determine and write down the gradient and y-intercept of the graph, that is the values and their units. Also write down the equation of the graph. Insert values for the gradient and the y-intercept. Include the units in brackets. This is the so-called empirical equation.
5.
 - a. Compare your empirical equation to equation (3) above. What seems to be the mathematical relationship between the force constant of the spring, k , and the gradient of the graph?
 - b. Calculate the force constant k , from the value of the gradient of the graph. Also give the units of the force constant.
6. Suspend the “unknown” mass from the spring and measure the elongation of the spring. Use equation (3) and the value of the force constant to determine its mass.

B. The elasticity of a rubber band

- ❖ Replace the spring with a rubber band. Repeat the table steps from Table 1 for the rubber band. Support the mass hanger in your one hand while you place the masses on the hanger with your other hand. Slowly lower your hand after placing each mass piece on the hanger. Wait until the rubber band stabilizes before taking a reading. When removing the mass pieces, it is advisable to grab the top mass and remove it slowly from the pile.
- ❖ Tabulate the values

7. Table 2

8. Draw the extension - load force graph of the data on the following page. Indicate on the graph in which direction the extension and contraction progressed.
9. Conclusion questions:
 - a) Are there any differences in the elasticity of the rubber band and the helical spring? If so, describe the differences in detail (there are at least three).
 - b) Do you think Hooke's law can be used to describe the elasticity for both rubber and steel (assuming that the different materials have different k values)? Explain.
 - c) If you were to design a spring scale, would you prefer to use a steel spring or a rubber band? Why?

Further Investigations

C.L. Stong. "Some Delightful Engines Driven by the Heating of Rubber Bands" Scientific American, April 1971

Safety precautions

Be careful with the spring and elastic. It could shoot you in the eye if it suddenly comes loose, especially when it is under tension.



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FSK 176

Practical Report

HEA: Heat

Objective

To investigate the relationship between the amount of heat added to a certain amount of water and the change in its temperature.

Study Aims

- Connecting a simple circuit containing a resistor and a switch.
- To measure the temperature of the water as it is heated by the resistor.
- To determine the amount of heat added to the water by the resistor.
- To establish the relationship between the temperature change of the water and the amount of heat added to it.
- To determine the specific heat capacity of water.

Introduction

You are going to use a resistor to heat the water in this experiment. An ammeter will measure the current, I , in the resistor. The current will be kept constant during the heating process. The current is the amount of electric charge, in coulombs, that flows through the resistor per unit time (seconds). It follows that the amount of charge that flows through the resistor in a time period t is

$$q(t) = It \dots\dots \text{constant current}$$

Electric energy, E , is transferred to the atoms in the resistor by the charge moving through it. This process is similar to that of a rope slipping through your hands while you are holding onto it. The effect is known as joule, ohmic, or resistive heating. The amount of energy, E , transferred to the resistor when an amount of charge q flowed through it is given by the electric potential difference, V (in volt), over the resistor.

$$V = \frac{E}{q} \dots\dots \text{constant voltage}$$

$$V = \frac{E}{It} \dots\dots \text{constant current and voltage}$$

$$E = VIt$$

This converted electric energy is transferred as heat, Q_R , by the resistor

$$Q_R = E$$

We will assume that all of the heat is transferred to the water

$$Q_W = Q_R = VIt.....(1)$$

This formula enables you to calculate the total amount of heat transferred to the water, Q_W , by the resistor after a time period t . The heat transferred to the water effects a change in its temperature, ΔT which you will calculate from measurements made with a thermometer.

Apparatus

- A polystyrene cup.
- A one ohm, 1Ω resistor with attached electrical leads to heat the water, and two clothes pegs to hold it in the cup.
- An alcohol thermometer with 0.5 °C divisions.
- A stopwatch for timing the heating process.
- A variable DC power supply that can deliver at least 3 A with internal volt and ammeters.
- A switch to switch to open and close the circuit.
- An electrical lead.
- An electronic scale to determine the mass of the water.

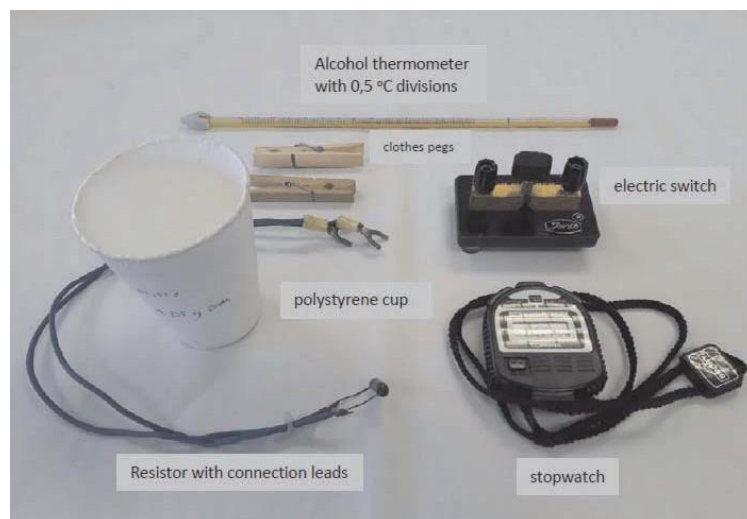
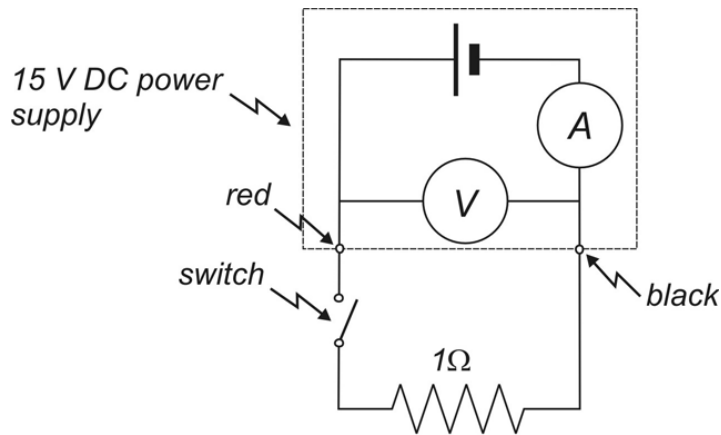


Fig. 1 The cup, resistor, thermometer, switch and stopwatch used in the experiment

Safety precautions

- Be careful not to break the thermometer. The resulting glass shards could cut you. Inform the demonstrator if you do break it.
- Be careful with the resistor. It gets quite hot and can burn you.



o - connecting terminals

Fig. 2 A circuit diagram showing the connections between the power supply, switch, and resistor.

1. Experimental Aim

Setup

- Check that the power supply is switched off and connect the resistor and (opened) switch in series with the power supply's terminals, see figure 2. Call the demonstrator to check your circuit.
- Fill the cup to about two thirds full with water and determine the mass, m_w , of the water.
- Put the resistor into the water and use the clothes pegs to suspend it in the water. Keep the resistor away from the sides and bottom of the cup otherwise it may burn through it! Be careful not to tip the cup over!
- Close the switch and adjust the power supply to deliver the required amount of current as instructed by the demonstrator.
- Note the current, I , in, and the potential difference, V , over, the resistor in your report book.
- Open the switch again

Measurements

First read through all of the steps of the following instructions before starting with the measurements. Make sure that you understand what you are supposed to do.

- a) Measure the initial temperature, T_o , of the water after stirring it a bit. Estimate the fractional value of the temperature of the water accurate to 1/10th of a degree.
- b) Make sure that all electrical connections are tight.
- c) Close the switch by pushing the peg in deeply (make a tight connection) and start the stopwatch.
- d) Keep on stirring the water while the resistor is heating the water. Record the temperature and time readings at one minute intervals for five minutes, again accurate to 1/10th of a degree.
- e) Again, take current, I , and potential difference, V , readings.
- f) Open the switch.

Results

2. Readings

Mass of water:

Current and potential difference readings:

3. Table:

- Calculate and tabulate the temperature change, ΔT , of the water with respect to the initial temperature, T_o .
- Calculate and tabulate the amount of heat, Q_w , that was added by the resistor to the water for each of your measurements, using **equation (1)**. Use the average current and potential difference values for the calculations.

4. **Graph:** Draw a graph that indicates the change in temperature of the water as a function of the amount of heat added to it.

Determine the values and units of the gradient and y-intercept of the graph.

5. **a)** Write the mathematical relationship between ΔT and Q_w down, according to your graph, with values and units (*supply the units in brackets for each of these*) for the constants in the relationship. This is the experimentally determined relationship and we call it the empirical relationship between the variables.

According to the theory the relationship between ΔT and Q_w should be

$$Q_w = m_w c_w \Delta T \dots\dots\dots (2)$$

where m_w is the mass of the water and c_w is the specific heat of the water. Make ΔT the subject of the equation, and then write your empirical formula down below it. Notice that both equations show a directly proportional relationship between ΔT and Q_w if we assume that the y-intercept of the graph is bought about by errors.

b) Compare the two formulas. Both formulas describe the same physical process and should actually be the same formula. How are the gradient of your graph and the $m_w c_w$ product then related if the formulas are the same?

c) Use only your experimental values of m_w and the gradient of your graph to calculate a value for the specific heat capacity of water, c_w , according to your answer in Q5b) above.

6. **a)** Calculate the percentage difference between your experimentally determined value for c_w (the calculated value in Q5c) and the accepted value of $4200 \text{ J/kg}\cdot^\circ\text{C}$.
- b) Conclusion question:** What factors could have led to the difference between the experimental and the theoretical specific heat capacity values for water?



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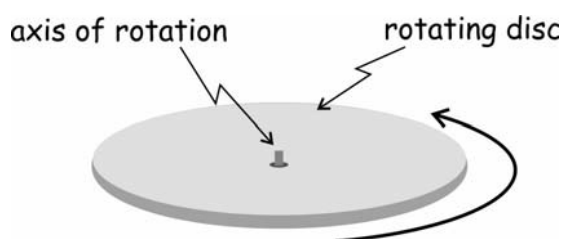
FSK 176

Practical Report

ROI: Rotational Inertia

Objective

To experimentally determine the rotational inertia, I , or moment of inertia, of a flat disc rotating about a perpendicular axis through its center.

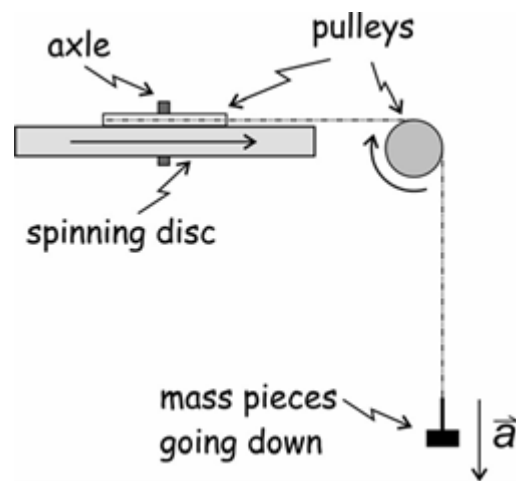


Study aims

- To apply a torque, τ_a (Greek lower case tau symbol), to a disc and to determine its value.
- To measure the resulting angular acceleration, α , of the disc as produced by the torque.
- To draw an appropriate graph from which the moment of inertia, I , of the disc can be determined.
- To use the above-mentioned graph to determine the opposing torque, τ_a applied to the disk by friction.

Introduction

In this experiment you will use the weight, W , of a hanger and some mass pieces on it to apply a torque, τ_a , to the disc. The hanger is attached to a piece of thread that passes over a pulley and is subsequently wrapped around another pulley mounted on top of the disc. The hanger accelerates downwards and creates a tension, T , in the thread which sets up the torque that spins the disc. You will vary the torque by altering the mass on the hanger and use a stopwatch to time the revolutions of the disc. From these measurements you can then calculate the applied torque and angular acceleration, α , of the disc.



Apparatus

- A base onto which the disc, with a pulley on top of it, is mounted horizontally with an axle - see **figure 1**.
- A spirit level for leveling the base and a short piece of wooden dowel.
- A vernier caliper for measuring the diameter of the pulley mounted on top of the disc.
- A ruler for measuring the diameter of the disc.
- Another separate pulley that can be mounted with a clamp on the edge of the laboratory work surface - the black pulley on the right-hand side of **figure 1**.
- A stopwatch for timing the disc's rotation.
- A mass hanger with some small mass pieces and some thread.
- An electronic scale to determine the mass hanging from the thread.

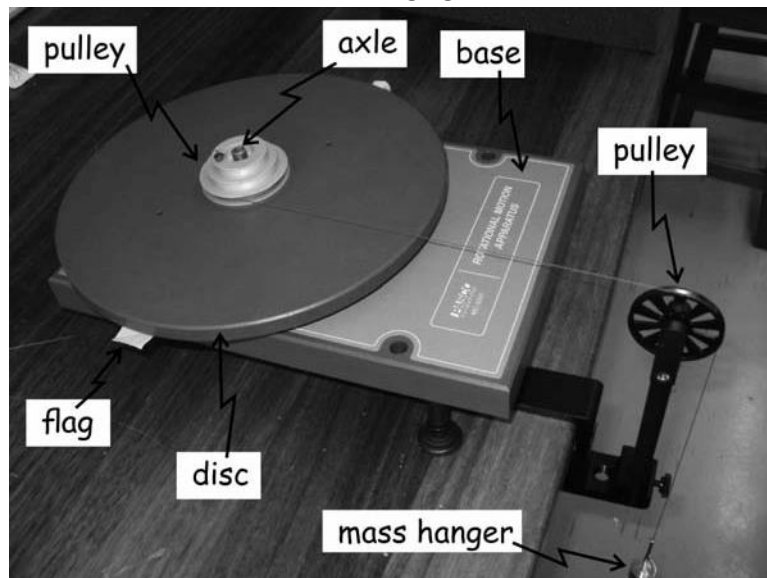


Figure1: The rotational inertia apparatus

Theory

Before the practical consult Physics textbooks and internet resources in order to better understand the following:

- How to measure the angular position, θ (in radians), of a rotating object
- Angular velocity, ω (in radians per second), of a rotating object
- Angular acceleration, α (in radians per second squared), of a rotating object
- Equations of motion for an object that is rotating with a constant angular acceleration
- What the rotational inertia or moment of inertia of a body measures
- Calculating the torque of a force acting on a body
- Newton's second law for rotation

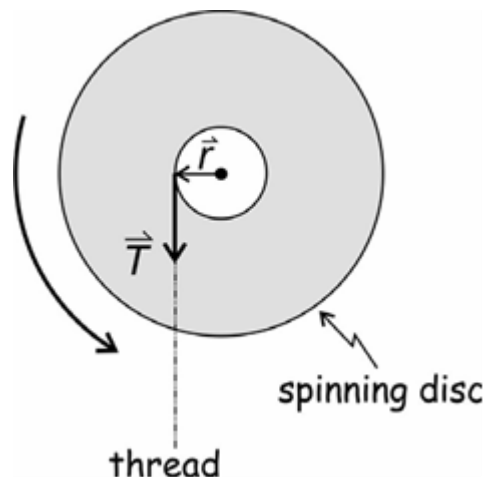


Figure 5: The applied torque on the disk

The tension, T , in the thread applies a torque, τ_a , to the disc according to

$$\tau_a = rT \quad \dots\dots\dots (1)$$

where r is the radius of the pulley mounted on top of the disc - **figure 2**. The same tension is applied to the hanging mass pieces of mass m , as they move with an acceleration $\vec{a}$ downwards - **figure 3**. Applying Newton's second law to the accelerating mass, we get

$$T = m (g - a) \quad \dots\dots\dots (2)$$

- see **figure 3**. The angular position, θ (in radians), of the disc is measured with respect to a line that remains fixed in a certain direction, line a in **figure 4**. The angular position of the disc is then the angle that a chosen line, rotating with the disc, line b in **figure 4**, makes with line a. The angular position of the disc continually increases as the disc spins around, since the angle between line b and line a increases with time. Note that the angular position does not revert back to zero when line b passes line a again.

If the angular acceleration of the disc, α , is constant, and the disc is accelerating from rest, and line b coincides with line a when it starts, then we can show from the rotational kinematics equations that the angular position, θ , of the disc after a time period, t , will be given by:

$$\begin{aligned} \theta &= \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2 \\ \therefore \theta &= \frac{1}{2} \alpha t^2 \quad \dots\dots\dots (3) \end{aligned}$$

The relationship between the net torque, τ_{net} , on the disc and its angular acceleration, α , is given by Newton's second law for rotation

$$\tau_{\text{net}} = I \alpha$$

where I is the moment of inertia of the disc about the rotation axis. If friction applies a constant opposing torque, τ_o , to the disc, this becomes

$$\tau_a - \tau_o = I \alpha$$

where τ_a is the applied torque. It follows that

$$\tau_a = I \alpha + \tau_o \quad \dots\dots\dots (4)$$

This is the linear relationship between the applied torque and the angular acceleration of the disc.

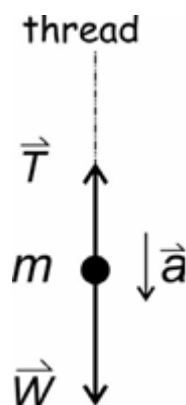


Figure 6: Mass pieces going down

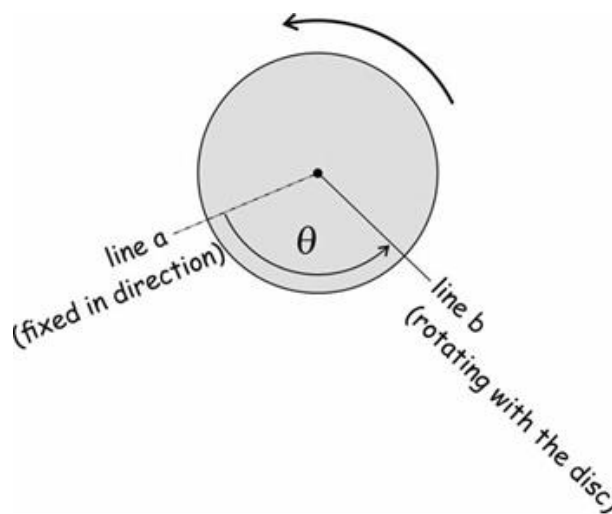


Figure 7: The angular position of the rotating disc

Setup

1. Clamp the black pulley with the mounting rod vertically onto the edge of the work surface.
2. Place the base just beyond the pulley on the work surface and level it with the spirit level by screwing its adjustable feet into or out of the base - see **figure 1**.
3. Determine the mass of the mass hanger, m_h , the mass of the disc, m_d , and the diameter of the disc, D_d . Cut a piece of thread that is long enough to reach from the center of the base to the floor when it passes over the black pulley. Tie one end to the mass hanger and the other end to the largest pulley on the disc.
4. Put the axle vertically in the bearing in the base and place the disc horizontally onto the axle.
5. Drape the thread over the black pulley and raise or lower the pulley in the clamp so that the thread runs parallel to the top surface of the disc. Change the angle of the black pulley so that it is parallel to the thread and not at an angle to it - see **figure 5**.
6. Keep the thread under tension while turning the disc by hand so that the thread winds onto the pulley on the disc. Stop when the hanger is just below the screw at the bottom of the clamp - see **figure 1**.
7. Hold the disc in this position and place the piece of dowel on the work surface just next to the flag on the disc.
8. Let go of the disc and count how many complete revolutions it makes before the hanger lands on the floor.

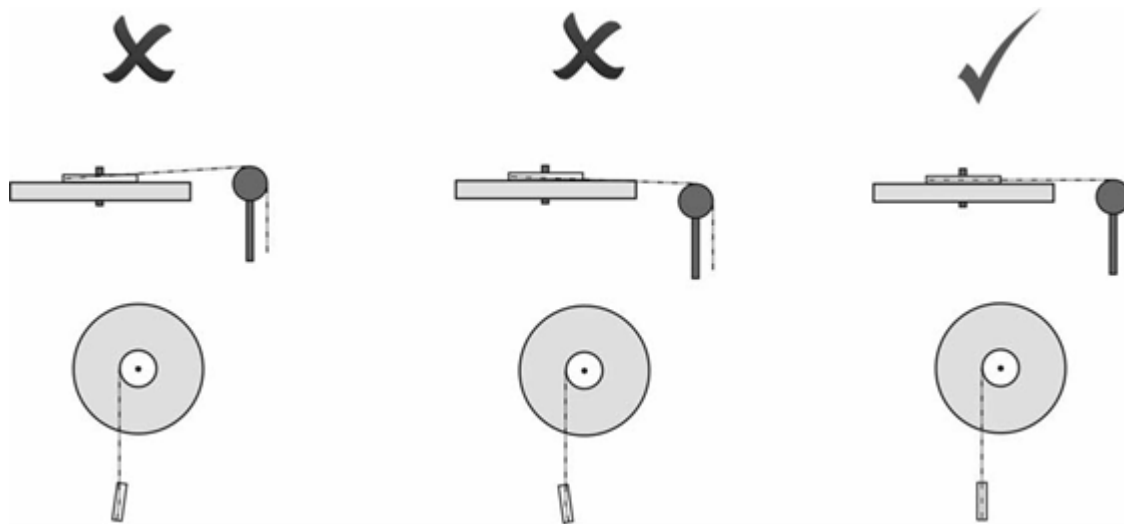


Figure 8: Adjusting the height (top 3 figures) and direction (bottom 3 figures) of the pulley on the rod for optimal conditions

Experiment

1. Experimental Aim:

Measurements

- Use the stopwatch to measure the elapsed time for the disc to complete a few revolutions, N , out of rest. Stop the stopwatch before the hanger reaches the floor.
- Repeat the measurement three more times and calculate the average time, t_{ave} . Keep the number of revolutions the same for each measurement.
- Make another four such measurements, each time with more mass on the hanger. Note that the angular position of the disc, θ in radians, increases by 2π radians for each revolution it completes.
- Tabulate your measurements and values.
- Also measure the diameter of the pulley on the disc, D_p , with the vernier caliper.

Data analysis

2. Table

Construct a table with columns for the accelerating mass, m , the average time, t_{ave} , the downwards acceleration, a , the tension in the string, T , the applied torque to the disc, τ_a in newton meters, and the angular acceleration, α in radians per second square, of the disc. Supply units for each of these variables and also a caption above the table.

3. a) In the space below, explain how you can use the time measurements, t_{ave} , to calculate the downward acceleration, a , of the mass pieces. Assume that the acceleration is constant.
 - ❖ THEN: Calculate and write the values of the downwards acceleration, a , of each of the mass pieces in the table.
- b) Explain how you can calculate the tension, T , in the thread while the mass pieces are going down.
 - ❖ THEN: Calculate and tabulate the tension in the string, T , for each of the mass pieces as they are going down.

The tension in the string is creating a turning effect on the disc - see **figure 2**. We call this the torque, τ_a , with units newton·meters.

- ❖ Calculate the applied torque on the disc - **equation (1)** - for each of the tensions in the string and fill the values in on your table.

This turning effect, or torque, causes an angular acceleration, α , of the disc which we observe as the disc spinning faster and faster as the mass pieces are going down. We are going to use **equation (3)** to calculate it.

$$\theta = \frac{1}{2}\alpha t^2 \quad \dots\dots\dots(3)$$

Here θ is the amount by which the angular position of the disc changed during the time period t . We are going to calculate the angular position of the disc from the start of its motion up to the measured time periods t_{ave} . That means we need to calculate by how much the angular position of the disc advanced during N revolutions. Since the angular position of the disc advances by 2π radians for each revolution, the angular position of the disc after a time period t_{ave} will be $N 2 \pi$ radians.

- ❖ Calculate the angular acceleration of the disc for each of the measurements and fill the values into the table.

4. **Graph:** Now draw a graph of the applied torque on the disc, τ_a , against its angular acceleration, α .
[Insert the graph on the next page.]

5. Determine the gradient, $grad$, and y-intercept, y_0 , of the graph. Supply values and units for these.

Note that radians is actually a dimensionless unit since it is calculated by a distance/distance or more precisely

$$\theta = \frac{s}{r}$$

where s is the arc length and r the radius of a circle – see

<http://www.mathsisfun.com/geometry/radians.html>

for more information. That means that it is only necessary to actually write the units *rad* to indicate that angles were measured in radians and not degrees, otherwise it can be omitted since it is a dimensionless quantity.

6. **a)** Write down the equation of your graph in the form

$$\tau_a = grad \alpha + y_0$$

here τ_a , and α are variables but $grad$ and y_0 are constants. Supply values and units (units in brackets) for these two constants in the equation of your graph. Expand the units of $grad$ into the basic S.I units in order to see if some of them do cancel out. This is called the empirical equation.

- b)** Compare the empirical equation to **equation (4)**. What seems to be the values and units of the moment of inertia I and the opposing torque, τ_0 ?

Conclusion

7. We used this experiment to measure the moment of inertia of the disc about a perpendicular axis through its center. For such a regularly shaped object we can derive a formula for its value

$$I = \frac{1}{2}MR^2 \dots\dots\dots (5)$$

where ***M*** is the mass of the disc and ***R*** its radius.

- a. Calculate the value of ***I*** using **equation (5)**.
- b. Compare $I_{\text{experimental}}$ from Q8 with the calculated value of ***I*** from Q9.1, and discuss your findings.



DEPARTMENT OF PHYSICS

FSK 176

Practical Report

SHM: Simple Harmonic Motion

Introduction

We find many examples of oscillating systems in everyday life. Watches use the precise oscillations of quartz crystals to keep time accurately. Information is encoded into the oscillations of electromagnetic waves and then transmitted to satellites which beam the signal back down to earth. Sound waves are propagated via oscillating air molecules.

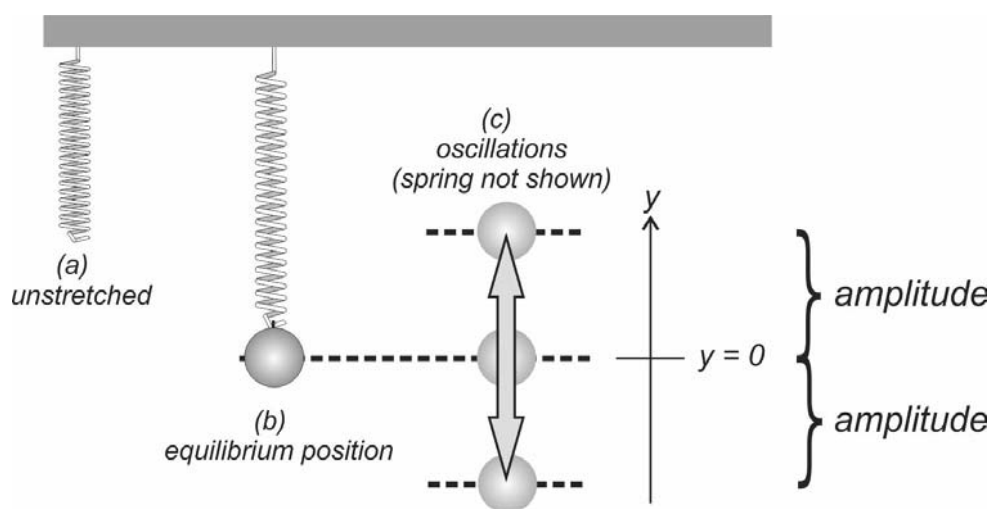


Figure 1: An oscillating mass on a spring.

In this experiment we will investigate the oscillations of a mass piece hanging on a spring, **figure 1**. The y -position of the mass piece is a function of time, $y_e(t)$ (e - experimentally determined), and we will calculate these positions from the restoring force of the spring, F_R , as measured by a force sensor connected to a computer - **figures 2 and 3**. We will then compare these positions to those calculated for simple harmonic motion, $y(t)$, a special category of oscillations. These two sets of y -positions should be very much the same if the oscillating mass piece exhibits simple harmonic motion.



Figure2: Spring hanging from the force sensor

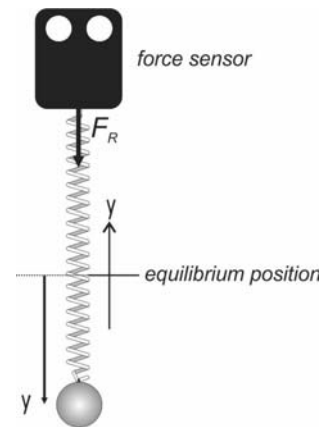


Figure3: The restoring force of the spring.

Theory

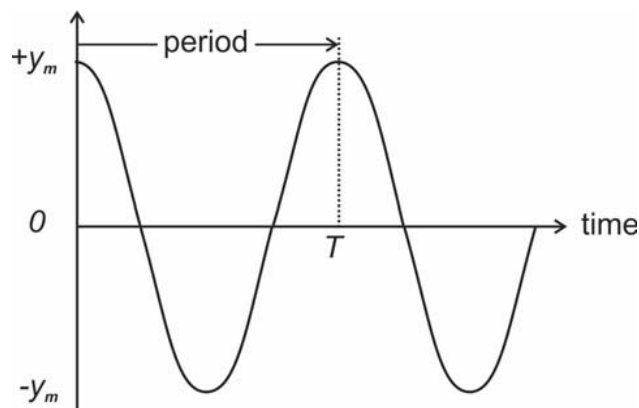
We will assume that the restoring force of the spring obeys Hooke's Law - see **figure 3**.

$$F_R = k y \quad \dots\dots\dots (1)$$

The force sensor will be zeroed with the mass piece hanging in the equilibrium position. That, together with the fact that it displays downward-acting forces as negative leads to the omission of the negative sign normally found in **equation 1** (the restoring force at the bottom end of the spring acts in the opposite direction to that which the force sensor measures at the top). The y -positions of the mass can then be calculated if the force constant, k , of the spring is known.

$$y_e = F_R / k \quad \dots\dots\dots (2)$$

The y -positions of simple harmonic motion are depicted in the graph below as a function of **time**.



y_m is the amplitude of the oscillations. We will use the extreme positions of the mass about the equilibrium position, $y = 0$, for the value of the amplitude - see **figure 1**. As you can see this graph is a basic cosine function

$$y = y_m \cos (\theta) \quad \dots\dots\dots (3)$$

θ is called the phase of the motion. It is time-dependent and is given by

$$\theta = \omega t \quad \dots\dots\dots (4)$$

where t is time and ω is the angular frequency. The angular frequency is a constant that determines the rate at which the oscillations occur and is measured in radians per second (see the section at the end of the notes for some information on the radian angular measure). The angular frequency, ω , is dependent on the period of the oscillations, T (the time to complete one cycle) - see the above graph. It is given by

$$\omega = \frac{2\pi}{T} \text{ rad/s} \dots\dots\dots (5)$$

It follows from **equation 4** that $\theta = 2\pi$ radians when $t = T$, that is, the phase of an oscillation or cycle is 2π radians and the y -position at this point is the same as that at $t = 0$ - see the graph above.

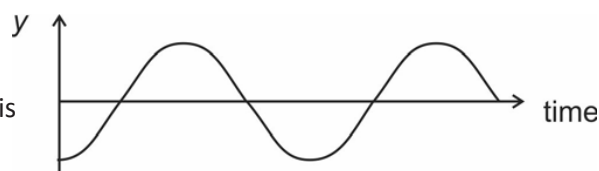
In general, we would not have that $y = +y_m$ at $t = 0$. Consider the following three cases of simple harmonic motion:

1. At $t = 0$, the y -position $-y_m$ in this case. **Equation 3** needs to be modified to describe this new graph below, and the way we do it is to add an angle ϕ called the phase angle or phase constant so that the phase is in general given by

$$\theta = \omega t + \phi \dots\dots\dots (6)$$

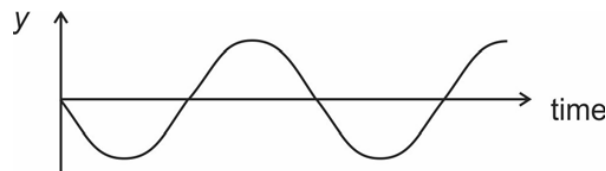
In this case $\phi = \pi$ radians so that the equation of the graph is

$$y = y_m \cos (\omega t + \pi)$$



2. Now we have that $y = 0$ at $t = 0$. It follows from **equations 6 and 3** that

$$\begin{aligned} \cos (\phi) &= 0 \\ \therefore \phi &= \pi/2 \text{ rad} \end{aligned}$$

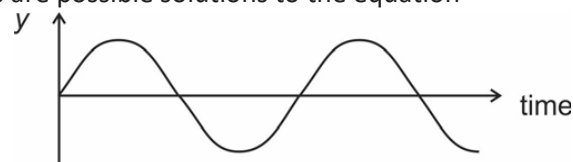


Check it with your calculator set to radians. We now have for this graph that

$$y = y_m \cos (\omega t + \pi/2)$$

3. Here we again have that $y = 0$ at $t = 0$, but now the graph continues along a different path - upwards instead of downwards as in the previous case. Note that due to the symmetry of the cosine function - see **figure 4** - that both the positive and the negative angles are possible solutions to the equation

$$\begin{aligned} \cos (\phi) &= 0 \\ \therefore \phi &= \pm \pi/2 \text{ rad} \end{aligned}$$



Try it with your calculator. So which solution is the correct one or will both do?

According to **equation 6** we have that at $t = 0$, $\theta = \phi$ and as time progresses the value of θ will steadily increase from this angle onwards. Looking at **figure 4** below (note the x-axis is θ), we see that if we were to choose the solution $\theta_0 = \phi = \pi/2 \text{ rad}$ (point a), then the graph of oscillations, would continue downwards. But if we take the solution $\theta_0 = \phi = -\pi/2 \text{ rad}$ (point b), then the graph will continue upwards. So, in case 3

$$y = y_m \cos (\omega t - \pi/2)$$

is the correct formula for the graph.

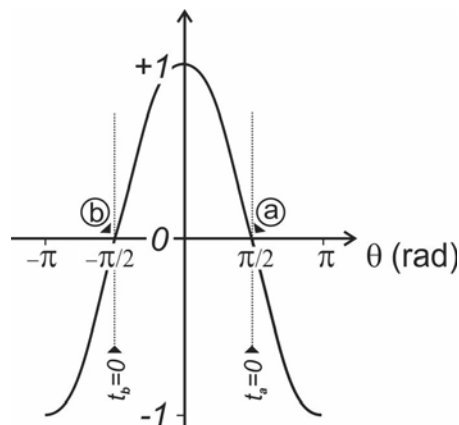


Figure 4: The cosine function

The y-displacement formula for simple harmonic motion is

$$y(t) = y_m \cos(\omega t + \phi) \quad \dots\dots\dots (7)$$

We will have to determine the values of y_m , ω and ϕ that go with the oscillating mass in order to calculate $y(t)$ from it.

When a spring drives the motion of the mass, with mass m , the simple harmonic theory indicates that the angular frequency, ω , can also be calculated from the force constant, k , of the spring according to

$$\omega = \frac{k}{m} \quad \dots\dots\dots (8)$$

More information can be found in your textbook in chapter 15.

Apparatus

A force sensor is used in this experiment to measure the restoring force that drives the motion of the mass piece - see **figure 2**. It uses a strain gauge to measure the force. The sensor connects to a computer via an interface box - see **figures 5 and 6**. It can measure a push or a pull acting on it up to 50 N. The sensor registers a downward pull as a negative force and an upward push on it as positive force. It can be zeroed by pushing the Tare button on the sensor. The Science Workshop program on the computer reads the sensor via the interface box when it is in either the **MON**itoring or **REC**ording mode - see **figure 7**.



Figure 5: The front of the computer interface.



Figure 6: The rear of the computer interface

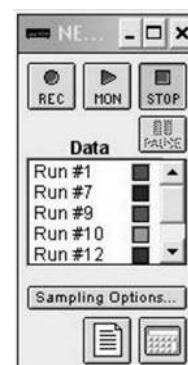
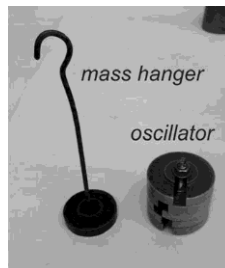
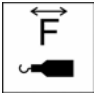


Figure 7: The program SetUp window

Apparatus setup

1. Mount the force sensor on the rod on the laboratory stand with its hook pointing downwards - see **figure 2**.
2. Plug the sensor into channel A of the interface box - **figure 5**. The indentation on the metal rim of the plug should be towards the *TOP*.
3. Check that the interface box is switched on and plugged in - **figures 5 and 6**.
4. Open the program file on the computer as instructed by the demonstrator. The program reads the sensor when it is put into either the monitoring (**MON**) or the recording (**REC**) mode - **figure 7**.
5. Click in the SetUp window to activate it and open the Sampling options. Set the periodic samples to 50 Hz and click OK. The program will read the force sensor 50x per second.
6. Maximize the SetUp window and double click on the force sensor icon below Analog Channel A. Set the Sensitivity to Med (10x) and click OK. Restore the SetUp window back down.
7. Hang the spring on the hook of the sensor and suspend the mass hanger from it with no mass pieces on it. Loosen and then tighten the thumb screw on top of the force sensor. Zero the sensor by pressing the Tare button on the sensor while the hanger is hanging still.



Experiment

1. a) Experimental Aim:

DETERMINING THE FORCE CONSTANT OF THE SPRING

Use the monitoring mode - **MON figure 7** - of the program to measure the restoring force while you stretch the spring downwards by a known amount. The force values appear in the Digits display window.



Figure 8: The Digits display window.

1. **b)** Calculate **k** in N/mm. Use the average of several measurements for your calculation. **STOP** the monitoring mode when you are done and remove the hanger from the spring.

Notes:

- Do not stretch the spring more than about 10 cm.
- Zero the sensor with the mass hanger hanging stationary.

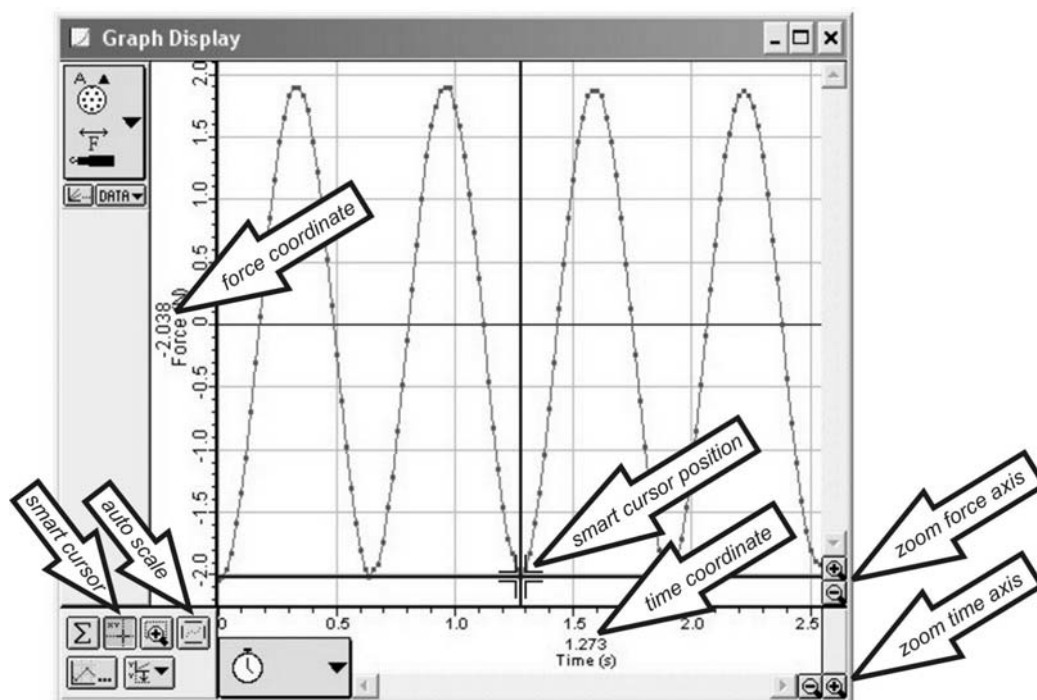


Figure 9: Recorded oscillations as displayed in the Graph Window of the program.

RECORDING SEVERAL OSCILLATIONS

Replace the mass hanger with the mass pieces that are fixed together - the oscillator. Record a few oscillations of the oscillator with the program. Call the demonstrator to check your recording before proceeding with the rest of the experiment.

Notes:

- Zero the sensor (Tare button) with the oscillator hanging stationary in the equilibrium position ($y = 0$).
- First set the oscillator in motion before starting the recording.
- Keep the amplitude of the oscillations small - a few centimeters at most.
- The oscillator should not swing sideways while you are recording its motion.
- Recordings can be deleted by clicking on their runs - **figure 7** - and pressing the Delete key on the keyboard.

THE Y-DISPLACEMENTS OF THE OSCILLATOR

Next you are going to calculate the experimentally-determined y -displacement, y_e , of the mass at several points in the **first recorded oscillation**.

Maximise the Graph Window and click the autoscale button in it - see **figure 9**. The graph displays how the restoring force of the spring, F_R (the dependent variable), changed with time, t (the independent variable).

Drag the cursor over a datapoint in the graph. The time and force coordinates of the chosen datapoint are selected in the table window - **figure 10**.

Select eight datapoints in the **first** oscillation in the graph. Try and space the chosen datapoints evenly throughout the oscillation. Calculate y_e in millimeters for each of the datapoints and tabulate your results.

[Note: Only populate the first three columns of the graph at this point (time, Force and y_e). The remaining two columns will be populated at questions 7 & 8]

Index	Run #1	Force (N)
t1	0.000	-2.045
1		
t2	0.020	-2.014
2		
t3	0.040	-1.953
3		
t4	0.060	-1.831
4		
t5	0.080	
5		
t6	0.100	
6		
t7	0.120	
7		
t8	0.140	
8		
t9	0.160	-0.305
9		
t10	0.180	0.061
10		

Figure 10: The Table window of the program.

2. Table

DETERMINING THE ANGULAR FREQUENCY AND AMPLITUDE

Click on the smart cursor in the Graph Window - **figure 9** - and move it around in the graph area. Note how the force and time coordinates of the cursor are displayed next to the Time(s) and Force (N) axis labels of the graphs - also indicated in **figure 9**.

3. a) Use the smart cursor to take readings on the graph to determine the period, T , of the oscillations. Take several readings and calculate an average period. Make a rough sketch of the graph and indicate on it where you made your readings.
- b) Calculate the angular frequency of the oscillations from **equations 5 and 8**. Note that you need to convert the force constant to S.I units. The values should be almost the same. Use the average of the two values for the y - displacement calculations you are going to make later on.
- c) Use the smart cursor to read off the minimum and maximum restoring force values, $F_{\min}$ and $F_{\max}$, and calculate the amplitude, y_m , of the oscillations in millimeters from these

$$y_m = \frac{F_{\max} - F_{\min}}{2k}$$

DETERMINING THE PHASE CONSTANT, ϕ , AND THE Y-DISPLACEMENTS OF THE SIMPLE HARMONIC OSCILLATOR

4. a) Again, use the smart cursor to read off F_0 , the restoring force of the spring when $t = 0$. Calculate y_0 , the displacement of the oscillator when $t = 0$ from it and subsequently determine the phase constant, ϕ (in radians) by setting $t = 0$ in **equation 7**. Note that you might need to adjust the obtained value of ϕ as explained in the theory section of these notes.
- b) Set up the simple harmonic motion y-displacement equation, **equation 7**, with values and units, in brackets, for the constants in the equation.
- c) Use this formula to calculate the y-displacements of the simple harmonic oscillator, **$y(t)$** , at the same stages as the eight y_e values you chose previously. Calculate and tabulate the $y(t)$ values, in millimeters, together with the difference between the two displacements: **$\Delta y = y_e - y$** .
[Tabulate the $y(t)$ values and the Δy values in the last two columns of the table used in Question 2]
- d) **Conclusion question:** Does it seem that the oscillator exhibited simple harmonic motion during the first recorded oscillation? Explain.

RADIAN ANGLE MEASURE

Just as we can measure distances in different measures like kilometers or miles, for example, the distance between Johannesburg and Pretoria is about 53 km or 33 miles, so angles can be measured in degrees or radians. Here we have that 360° is 2π radians, so that 40° is converted to radians as follows:

$$360^\circ = 2\pi \text{ rad}$$

$$1 \text{ rad} = 1 \text{ rad} \times \frac{180^\circ}{\pi \text{ rad}} = 57.3^\circ$$

$$\therefore 40^\circ = 40^\circ \times \frac{2\pi \text{ rad}}{360^\circ} = \frac{2\pi}{9} \text{ rad} = 0.698 \text{ rad}$$

See <http://www.mathsisfun.com/geometry/radians.html> for more information.

SAFETY PRECAUTIONS

- Be careful of the supporting rod on the laboratory stand. It could poke you in the eye. Especially when it is pointing towards you.
- Be careful with the spring. It could shoot you in the eye if it accidentally comes undone.

APPENDICES

Appendix A – The Vernier Caliper

Appendix B – Electronic Scale

Appendix C - Stopwatch

Appendix A

The vernier caliper

The vernier caliper is an instrument that can make very accurate linear measurements. It was introduced in 1631 by Pierre Vernier of France. It utilises two graduated scales: a main scale similar to that found on a ruler and another graduated auxiliary scale, the vernier scale, that slides parallel to the main scale and enables readings to be made to a fraction of a division on the main scale. The webpage

<http://www.phy.uct.ac.za/courses/phylab1/vernier>

gives more information about the use of the vernier caliper.

The vernier caliper has two sets of jaws, one for measuring outer measurements (like the outer diameter of a disc) and one for inner measurements (like the inside diameter of a pipe). It also has a pin for measuring depths (like the depth of a groove) - see **figure 1**.

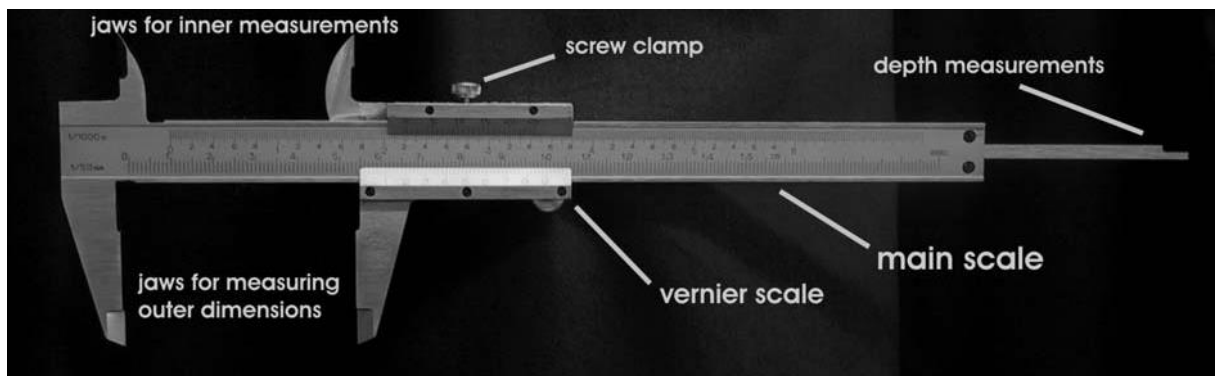


Fig. 1 a Vernier Caliper.

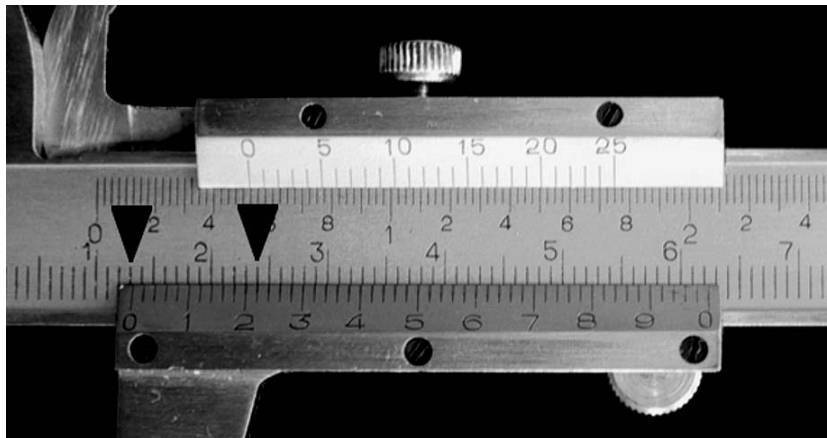


Fig. 2 Taking a reading on a vernier caliper.

The caliper in **figure 2** has 50 divisions on the vernier scale. The leftmost one, the index line, indicates the reading on the main scale, 13 mm in this case since the index line is just to the right of it. The reading on the vernier scale is taken at the position where a vernier line lines up with a line on the main scale. This is at the line just to the right of the 2 line on the vernier scale. This corresponds to a vernier reading of 0,22 mm. The caliper reading is therefore 13,22 mm (for all three types of measurements - outer, inner and depth measurements).

The meaning of the vernier divisions can be inferred by closing the caliper completely and noting that the 50 divisions on it are 49 mm long. One vernier division is therefore 0,98 mm wide. The first vernier division is therefore 0,02 mm short of the first millimeter mark on the main scale, while the second one is 0,04 mm short of the second millimeter mark on the main scale. If the jaws were therefore to be opened 0,02 mm the first vernier line will coincide with first millimeter line on the main scale, and if it were to be opened 0,04 mm the second vernier line will coincide with the second line on the main scale...

In **figure 3** the outer diameter of a cylinder is measured. The jaws were closed on it and the clamping screw tightened.

The smallest division on the vernier in **figure 3** is 0,02mm shorter than one millimeter.

The reading on the main scale  is 7mm

The reading on the vernier scale  is 0,28mm

The diameter of the cylinder is therefore 7,28mm

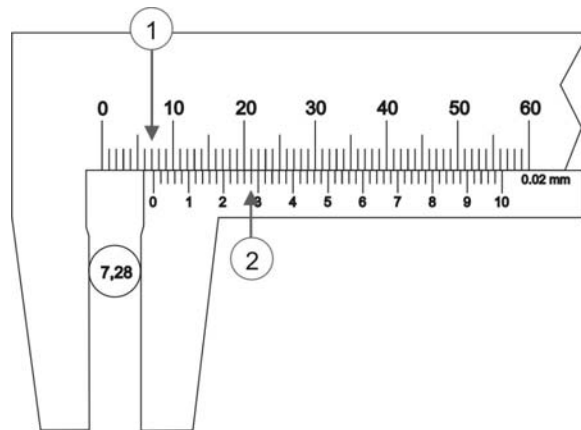


Fig. 3 Measuring with a vernier caliper

In order to determine the diameter it is therefore necessary to take readings on both the main- and vernier scales and combine these two readings. The vernier divisions of the caliper in **figure 4** are 0,05 mm shorter than two

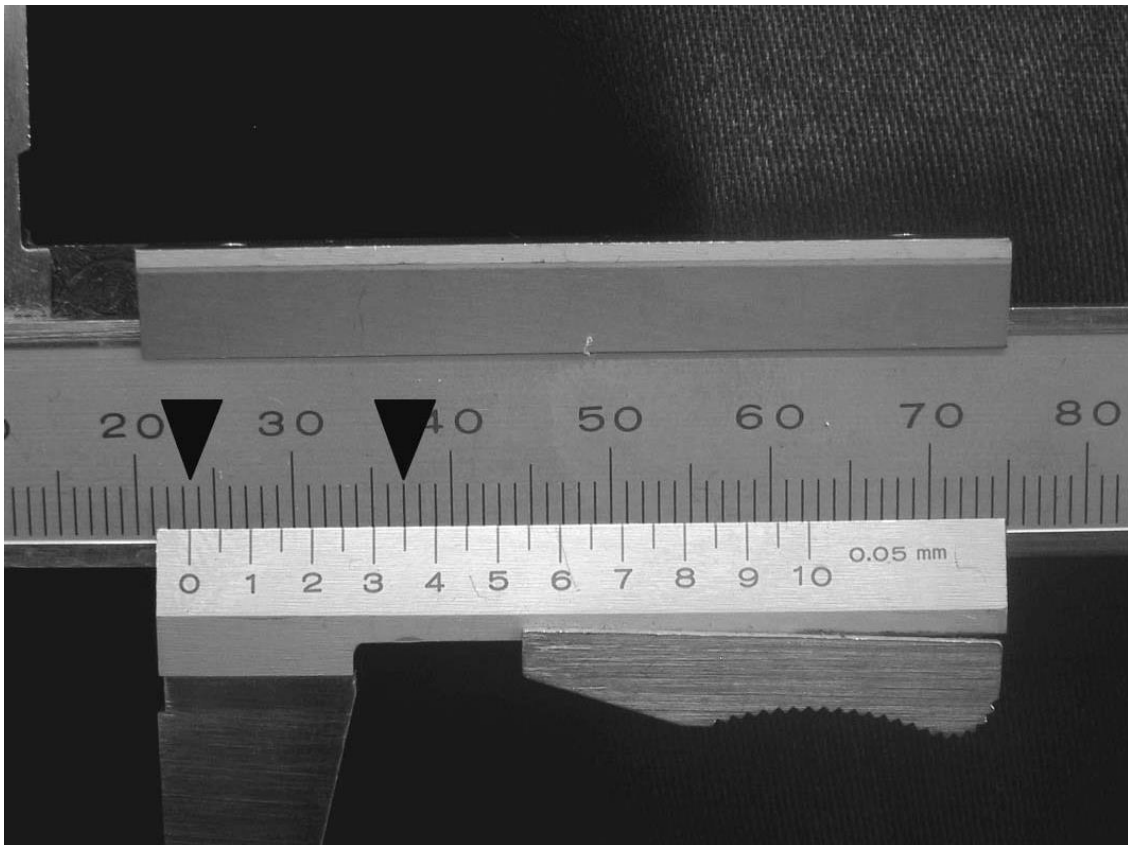


Fig. 4 Taking a reading on a vernier caliper.

millimeters. The reading on the main scale is 23 mm, while the vernier reading is 0,35 mm. The caliper reading is therefore 23,35 mm.

In **figure 5** another measurement is made with a similar caliper:

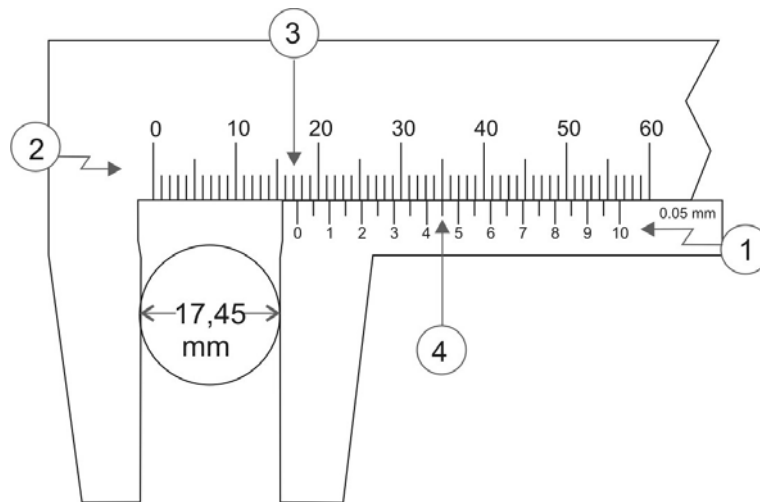


Fig. 5 Reading of a vernier caliper

The vernier $\blacktriangle$ is the auxiliary scale that slides along the main scale $\odot$.

The reading on the main scale is taken just to the left of the zero on the vernier scale $\blacktriangle$ (17mm in this case).

The reading on the vernier is taken where the marks on the vernier scale and the main scale coincide $\bullet$ (0.45mm in this case).

The final reading is the sum of readings on the vernier and the main scale.

The diameter of the object is therefore 17.45mm.

Appendix B

Electronic scale

METTLER TOLEDO PB1502 electronic scale



Fig. 1 METTLER electronic scale.

- Check that the security cable (**2** in **figure 1**) is not touching the scale pan and that the scale is not standing on the cable.
 - Ensure that the scale is level (**1**). If not, screw the feet below the front of the scale in (anticlockwise) or out (clockwise).
 - The scale is switched on and zeroed by pressing the **On** switch (**3**).
 - Place the object (< 1,5 kg) on the scale pan and read the mass in grams.
 - Note that the scale is sensitive to draughts due to the large scale pan (Bernoulli effect).
-

OHAUS electronic scale



Fig. 2 The OHAUS electronic scale.

- The maximum capacity of this scale is 200 grams. The scale displays an E symbol if this limit is exceeded. In such a case remove the load immediately from the scale pan.
 - Press the on/off button to switch the scale on.
 - The scale reading is zeroed by pressing the same button.
 - The weighing units can be changed by pressing the unit/cal button.
 - A star symbol, ×, appears on the left of the display when the reading is stable.
 - Hold the on/off button for 3 seconds to switch the scale off.
-
-

Appendix C

Stopwatch



Fig. 1 A SANJI stopwatch.

There are several types of digital stopwatches in the Physics laboratory, but most of them are operated in the same way.

The first step is to get it in the stopwatch mode, since it can be in various modes like time (watch), altering the time and setting the alarm. Use the mode button (**M** button in the figures) to change it to the stopwatch mode. In the stopwatch mode you will see a line of segments at the top of the display flashing while the digits are not flashing. Once in this mode you can reset the stopwatch to zero with the Reset button (**R** in the figures). The stopwatch is started and stopped with the **S** button.

12:34⁵⁶

Fig. 2 An example of a reading on a stopwatch.

Figure 2 shows an example of a reading on a stopwatch. The elapsed minutes appear to the left of the colon character (:). To the right of the colon is the seconds. The smaller digits are the decimal seconds. The reading is therefore 12 minutes and 34.56 seconds.

Please note that it is a good practice to use the index finger rather than the thumb to press the start button when taking measurements, since it is generally the more agile one of the two.

It is possible to “freeze” the display while the stopwatch keeps on running in the background. This is achieved by pressing the **LAP** or **SPLIT** button (**R** in **figures 1** and **3**). The normal mode of the stopwatch is restored by pressing this button again.



Fig. 3 Another SANJI stopwatch

